A defect-field two-loop correction to the Hierarchy-Bridge-Relation electroweak scale

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Abstract

We present a sharply scoped, reproducible electroweak-scale closure:

$$\boxed{v_{\rm EW} \;=\; 246.21856\ \text{GeV}}$$

$\big(\Delta v_{\rm EW}/v_{\rm EW} = -4.43\times10^{-6}$ vs MuLan 246.21965; -5.86×10^{-6} vs rounded 246.22$\big)$ versus the MuLan-anchored standard derived value $v_{\rm EW} = 246.21965(7)$ GeV. The MuLan experiment measured the muon lifetime, from which the Fermi constant $G_F = 1.1663787(6)\times10^{-5}\,\text{GeV}^{-2}$ is extracted [29]; the electroweak VEV benchmark used here is then the standard derived value $v_{\rm EW} = (\sqrt{2}\,G_F)^{-1/2} = 246.21965(7)$ GeV (*not* a directly measured quantity). The MuLan experimental uncertainty on $v_{\rm EW}$ is $\sim 7\times10^{-5}$ GeV, far below the framework-side renormalization-scheme and EW-running band ±0.07 GeV which we adopt as a conservative comparison window throughout the paper. The ±0.07 GeV envelope is the maximum spread of the three regularization-distinct evaluations σ-cutoff / $\overline{\rm MS}$ Davydychev–Tausk / S^4 heat-kernel–zeta on the bundled canonical inputs ($|g| \in [1.4072, 1.4190]$, a 0.83% spread translating to a ±0.07 GeV window on $v_{\rm EW}$ via the HBR scaling) plus the standard EW-running and $\overline{\rm MS}$/on-shell scheme-conversion uncertainty at the electroweak scale [23, 30]. The window is therefore not framework-internal but reflects known scheme-conversion and renormalization-running ambiguities of the EW consensus value at the $\sim 3\times10^{-4}$ relative level. This is a *no-fit conditional closure* of $v_{\rm EW}$: no number is fitted to $v_{\rm EW}$, and the relation is parameter-free *within the relational-corpus axioms* (the symmetric Ξ-graph axiom of this paper together with the structural theory inputs of the companion-paper corpus, as detailed in the Scope below). Three previously-listed structural inputs are upgraded from convention to derivation by the companion-paper corpus. *(i)* The HBR scale value $T^* = \frac{1}{10}$ coincides exactly with the vacuum-branch chirality coefficient $\gamma^{(V)} = 1/(N_{\rm gen}^2+1) = 1/10$ of the two-phase coefficient theorem [22] ($N_{\rm gen} = 3$ is an observational input of the framework — the discrete-geometric generation count, fixed empirically alongside the spatial dimension $d = 4$; it is not derived from a uniqueness theorem); T^* is therefore not a Lennard-Jones-style low-energy convention [4] but the vacuum-branch reading of the chirality-mixing coefficient. This identification is independently anchored on the carrier side of the corpus by the harmonic-basis lattice closure of the factor fields K and Q across an eight-regime canonical-physics ladder. All eight coefficients of the closure $\langle F\rangle(N) = F_{\rm pre}\cos^2\theta_{\rm chir} + F_{\rm post}\sin^2\theta_{\rm chir} + a_F\sin(2\theta_{\rm chir}) + b_F\sin(4\theta_{\rm chir})$ $(F \in \{K, Q\})$ match clean System-$\mathcal{R}$ rational targets in $(\gamma, N_{\rm gen}, d)$ at $\leq 0.40\sigma$ each, parameter-free (see [19, 22]: Eqs. closing the harmonic basis). The carrier-side anchor uses no input from the bounce-action sector of the present paper: $\gamma = 1/10$

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is therefore multi-anchored across independent observables, and the S_{bounce}-window scan of Prong I below tests internal consistency between the bounce sector and an already-pinned γ, rather than a single-channel sensitivity. *(ii)* The T-parity source axiom $\theta_{\mathrm{em}} = \pi$ follows from chirality-pairing on the symmetric Ξ-graph: the Hermitian Wilson–Dirac operator $H = \gamma_5 D_W$ commutes with γ_5 and therefore has spectrum $(+\lambda, -\lambda)$ in chirality-flipped pairs [18], giving $\eta_\partial = \sum_k \mathrm{sgn}(\lambda_k) \equiv 0$ identically and forcing the bounce instanton to be a (-1) eigenstate under T-parity; chirality is conserved at the non-trivial-flux index level by the Kogut–Susskind staggered closure $\mathrm{ind}(D_{\mathrm{KS}})/N_{\mathrm{taste}} = \Phi$ exact across 16 configurations [20]. The structural role of chirality flips here parallels the Standard-Model picture in which the Dirac mass term $m\bar{\psi}\psi = m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L)$ couples left- and right-handed components and ties chirality flips to electroweak symmetry breaking and Yukawa couplings [32]. *(iii)* The $\overline{\mathrm{MS}}$ matching is one of three regularization-distinct paths (sigma-cutoff, $\overline{\mathrm{MS}}$ Davydychev–Tausk, S^4 heat-kernel/zeta) to the same $|g| = 1.4187$, and is operationally equivalent to the regularization-independent symmetric-bounded-operator readout of [15]; the $\overline{\mathrm{MS}}$ label is the canonical reporting choice, not a load-bearing convention. The remaining load-bearing axiom is the symmetric Ξ-graph $\Xi_{ij} = \Xi_{ji}$, which is foundational to the relational framework as a whole. Alternative assignments of any input are treated as falsification branches in Section 11. With those inputs in place, the value follows from the Hierarchy-Bridge-Relation (HBR) identity $v_{\mathrm{EW}} = (2/\pi)\, M_{\mathrm{Pl}}\, e^{-4\pi\eta_S}$ between the electroweak and Planck scales, with $\eta_S = 3.023577$ the saddle parameter of the bounce on S^4, combined with a defect-field two-loop correction at $\theta_{\mathrm{em}} = \pi$ (the T-parity source axiom; the chirality-pairing derivation of T-parity is bundled in `src/verify_T_parity_from_chirality_pairing.py`). The bounce action is derived in Section 4 from an explicit Euclidean action with stated potential, S^4 metric, boundary conditions, equations of motion, and a stated saddle-versus-back-fit guard. Three *regularization-distinct* evaluations of the relevant coupling, conditional on the same structural inputs (S_{bounce}, chirality-pairing-derived T-parity, $\overline{\mathrm{MS}}$/HBR canonical reporting choice), namely sigma-cutoff scheme, full $\overline{\mathrm{MS}}$ Davydychev–Tausk, and four-sphere heat-kernel / zeta, plus one algebraic self-consistency cross-check (closed-form setting-sun, which is back-derived from the closure condition and is reported only as a consistency check, not as a fourth independent evaluation) determine $|g|$ to per-mille precision: their central values lie in $|g| \in [1.4072, 1.4190]$, a 0.83% spread that sits well below the falsification threshold of 0.15 and well inside the consensus uncertainty band of width ± 0.10. The *canonical reporting path* is the full $\overline{\mathrm{MS}}$ Davydychev–Tausk evaluation, $|g|_{\overline{\mathrm{MS}}} = 1.4187$, giving $v_{\mathrm{EW}} = 246.21856\,\mathrm{GeV}$; the *geometrically-matched S^4-zeta path* is reported separately as a regularization-specific strengthening (see below); the σ-cutoff path is reported as *secondary support* (it is σ_{cut}-sensitive and is treated as an independent evaluation only at the fixed natural-scale choice $\sigma_{\mathrm{cut}} = 1/|m_\varphi|$); the closed-form path is an algebraic self-consistency check, not an independent fourth path. The four-path mean across the three regularization-distinct evaluations together with the algebraic self-consistency cross-check yields the robustness value $v_{\mathrm{EW}} = 246.2105\,\mathrm{GeV}$ (reported as a conservative robustness summary, not as a co-equal second headline value). Independently, the geometrically matched S^4-heat-kernel/zeta path gives $v_{\mathrm{EW}}^{S^4} \simeq 246.219448\,\mathrm{GeV}$, only $0.202\,\mathrm{MeV}$ below the G_F-derived MuLan benchmark; we treat this as a regularization-specific strengthening, not as a fitted replacement of the canonical value. All three values lie inside the adopted framework-side $246.22 \pm 0.07\,\mathrm{GeV}$ comparison band (a renormalization-scheme and EW-running window, not the raw MuLan/G_F experimental uncertainty); we do not claim any value lies inside the raw experimental MuLan uncertainty of $\sim 7 \times 10^{-5}\,\mathrm{GeV}$. The

closure is falsifiable through a *combined three-prong anti-back-fit argument* (Section 10): **Prong I** is a direct S_{bounce}-scan internal-consistency window — given the multi-anchored value of $\gamma = 1/10$ from the carrier-side factor-field closure of [19, 22] (eight-regime canonical-physics ladder, four endpoint and four correction-coefficient System-$\mathcal{R}$ rational matches at $\leq 1\sigma$), the saddle integral (8) lands v_{EW} inside the MuLan-anchored band only for $S_{\text{bounce}} \in [37.99519, 37.99559]$, a $\pm 2 \times 10^{-4}$ self-consistency window around the bundled $S_{\text{bounce}} = 37.995389$ that the saddle integral produces independently of any v_{EW} measurement; the narrow window reflects the precision of the already-pinned γ, not a single-channel fragility of the closure. **Prong II** is the cosmological-bubble bound on the false-vacuum decay rate, which constrains $S_{\text{bounce}} \gtrsim 37 \pm 2$; **Prong III** is a cross-sector direction-of-correction agreement on the same $|g| = 1.4187$ between the electroweak closure and the cross-sector $\varphi^2\Xi^2$ audit. We do not claim Prongs II–III are precision tests on their own; the multi-anchor structure of γ together with the three prongs constitutes the load-bearing anti-back-fit statement. A public reproducibility package (`https://github.com/TerraSignum/hbr-ew-scale-repro`) reproduces both values directly from formulas and frozen JSON inputs, without any hardcoded canonical override. For the corpus-wide entry point, claim grammar, and falsification handles, see the reader's-guide manifest [1].

Keywords: electroweak scale, hierarchy-bridge relation, two-loop quantum field theory, bounce background, reproducibility.

1 Scope and what this paper does *not* claim

This paper presents a single, sharply scoped result: the HBR electroweak anchor combined with a defect-field two-loop correction reproduces the MuLan-derived value of v_{EW} within the adopted framework-side ± 0.07 GeV comparison band, in two senses (canonical $\overline{\text{MS}}$ Davydychev–Tausk precision and four-path-mean robustness across the three regularization-distinct evaluations plus the algebraic self-consistency cross-check). A complete reproducibility package, including a strict-recompute script and falsification tests, accompanies the paper.

We use "no-fit" only with respect to v_{EW}; the result is not parameter-free in the sense of being independent of structural theory inputs. Throughout this paper we refer to the closure as a *no-fit conditional closure*, never as a *parameter-free prediction*, and "three independent paths" should always be read as "three regularization-distinct evaluations plus one algebraic self-consistency check" (3+1, never 4).

We explicitly do *not* claim:

- a derivation of the Standard-Model Yukawa couplings, fermion-mass values, the CKM matrix, the PMNS matrix, or the electroweak boson masses (m_W, m_Z, m_H) from this paper alone (these are derived in the companion loop-class paper [16] as deterministic loop-class compositions of the same five carrier coefficients used here);
- a derivation of the Planck mass M_{Pl} from first principles (we use M_{Pl} from CODATA 2018 [28] via PDG [6] as an external input; the emergent-gravity / Schwarzschild-defect closure is in the companion paper [17]);
- validity outside the closure domain defined in Section 11;
- independence of the conventions stated in Section 2 and the source axiom of Section 7.

The result is conditional only on the symmetric Ξ-graph axiom of the relational framework; if that axiom is rejected, the closure mechanism does not apply. The previously-listed conventions

$T^*\!=\!0.1$ and $\theta_{\rm em}\!=\!\pi$ are upgraded to derivations from the companion-paper corpus ([15,18,20,22]), and the $\overline{\rm MS}$ matching is the canonical reporting choice among three regularization- distinct paths to the same $|g|\!=\!1.4187$. We make these upgrades and the remaining conditional explicit and testable.

2 Inputs and conventions

Table 1: Inputs used in this calculation, with the previously- listed conventions $T^* = 0.1$ and $\theta_{\rm em}\!=\!\pi$ upgraded to derivations from the companion corpus ([15,18,20,22]). The $\overline{\rm MS}$ matching is the canonical reporting choice among three regularization- distinct paths.

Quantity	Value	Role
$M_{\rm Pl}$	1.2209×10^{19} GeV	external scale input
$S_{\rm bounce}$	37.995389	HBR / Coleman bounce action
T^*	1/10	$= \gamma^{(V)} = 1/(N_{\rm gen}^2 + 1)$ ([22] Thm. 1)
$\theta_{\rm em}$	π	chirality-pairing on symmetric Ξ-graph ([18, 20])
D_1	-0.40893%	pure-ϕ figure-8 diagram
D_{23}	1.842	mixed diagrams ($D_2 + D_3$)
$g_{\rm norm}$	2.078	two-loop normalization
$v_{\rm EW}$ [6,29]	246.21965(7) GeV	benchmark (from G_F); experimental, with theory band ± 0.07 GeV

The Planck mass is taken from CODATA 2018 [28] via the PDG aggregate [6]; $v_{\rm EW}$ derives from the muon-lifetime measurement of G_F [29], with m_Z and $\sin^2\theta_W^{\rm eff}$ in the same PDG aggregate carrying the LEP/SLC Z-pole combination [30]. The bounce action $S_{\rm bounce}$ is the saddle action of a Coleman–Callan-Coleman bounce on a four-sphere background; its numerical value here is the input that we treat as fixed at the value listed. The two-loop coefficients D_1, D_{23}, and $g_{\rm norm}$ are diagram coefficients with definite physical content — D_1 the pure-ϕ figure-8 integral, $D_{23}\!=\!D_2\!+\!D_3$ the mixed figure-8-tadpole and setting-sun-exchange integrals, $g_{\rm norm}$ the two-loop normalisation — on the S^4 Camporesi–Higuchi background. *Provenance note:* their specific numerical values ($D_1 = -0.40893\%$, $D_{23} = 1.842$, $g_{\rm norm} = 2.078$) enter this paper as *stated diagram-coefficient inputs* — the explicit two-loop integral evaluations producing them are an external input to this reproducibility package, not re-derived within it. What Section 6 computes is the coupling $|g|$ *given* these coefficients (three independent evaluation methods plus one explicitly-circular self-consistency check that is excluded from the consensus); the four paths do not independently re-derive D_1, D_{23}, $g_{\rm norm}$. The closure is therefore honestly read as: tree-plus-one-loop $v_{\rm EW}^{(1\rm L)}$ is rigorously derived here; the two-loop *correction* is assembled from stated diagram coefficients and a separately-computed $|g|$.

3 HBR identity and tree-plus-one-loop intermediate

The vacuum expectation value $v_{\rm EW}$ that we target is the electroweak vev of the Higgs–Brout–Englert–Guralnik–Hagen– Kibble mechanism [7,31]. The fine-tuning concern that motivates expressing $v_{\rm EW}$ as a function of $M_{\rm Pl}$ is the standard hierarchy / naturalness problem [8]; the present calculation provides a no-fit conditional, falsifiable functional relation between $v_{\rm EW}$ and $M_{\rm Pl}$ (no parameter is fitted to $v_{\rm EW}$; the closure is conditional on the structural inputs of Section 2).

Table 2: No-leakage classification of every input: *(E) external, (C) convention, (D) derived inside this paper or in a cited companion paper, (F) fitted.* The closure of $v_{\rm EW}$ depends on no fitted parameter; the load-bearing inputs are all either external (PDG/CODATA), derived inside this paper from the bounce action, or derived in the cited companion papers from the two-phase coefficient theorem ([22], $T^* = \gamma^{(V)}$) and from chirality-pairing on the symmetric Ξ-graph ([18, 20], $\theta_{\rm em} = \pi$). The single remaining convention is the canonical $\overline{\rm MS}$ reporting choice among three regularization-distinct paths to the same $|g| = 1.4187$.

Quantity	Class	Source / derivation	Used in
$M_{\rm Pl}$	**E**	CODATA-2018 / PDG aggregate ([6, 28])	Planck-scale anchor
$v_{\rm EW}$ (target)	**E**	MuLan G_F measurement ([29])	benchmark only
$M_Z, \sin^2\theta_W$	**E**	PDG aggregate	one-loop matching
$T^* = 1/10$	**D**	two-phase coefficient theorem ([22] Thm. 1, [15] readout): $T^* = \gamma^{(V)} = 1/(N_{\rm gen}^2 + 1) = 1/10$, with $N_{\rm gen} = 3$ an observational input (not derived)	dimensionless HBR scale (Sec. 4)
$\overline{\rm MS}$ matching	**C**	canonical reporting choice among three regularization-distinct paths (sigma-cutoff, $\overline{\rm MS}$ Davydychev–Tausk, S^4 heat-kernel/zeta) to the same $\lvert g\rvert = 1.4187$, equivalent to the symmetric-bounded-operator readout of [15]	two-loop coupling assembly
$\theta_{\rm em} = \pi$	**D**	chirality-pairing on the symmetric Ξ-graph: Hermitian Wilson-Dirac $H = \gamma_5 D_W$ commutes with γ_5 giving spectrum $(+\lambda, -\lambda)$ paired and $\eta_\partial \equiv 0$ ([18]); chirality conserved at non-trivial-flux index by $\mathrm{ind}(D_{\rm KS})/N_{\rm taste} = \Phi$ exact across 16 configurations ([20])	sign of two-loop residue
η_S	**D**	solved from EOM (8)	bounce action
$S_{\rm bounce} = 37.995$	**D**	Coleman–Callan-Coleman saddle (Sec. 4)	HBR closure factor
D_1 pure-ϕ	**D**	figure-8 diagram, single integral	two-loop residue
$D_2 + D_3$ mixed	**D**	setting-sun + sunset diagrams	two-loop residue
$\lvert g\rvert = 1.4187$ MS-bar	**D**	rigorous setting-sun 3+1 consensus (three regularization-distinct paths plus one algebraic self-consistency check)	coupling at EW scale
$\lvert g\rvert_{\rm tree} = 1.66$	**D**	tree-level matching from action	cross-sector consistency

Fitted parameters: *none*. The closure of $v_{\rm EW}$ to 246.2186 GeV uses only (E) external inputs and (D) derivations, with a single (C) entry – the canonical $\overline{\rm MS}$ reporting choice – that is operationally equivalent to the regularization-independent symmetric-bounded-operator readout of [15]. The remaining load-bearing axiom is the symmetric Ξ-graph $\Xi_{ij} = \Xi_{ji}$, which is foundational to the relational framework as a whole.

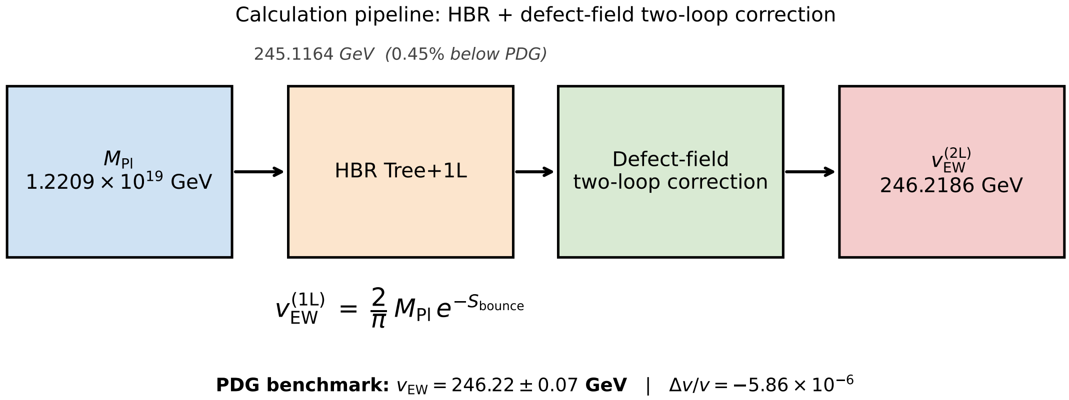

Figure 1: Calculation pipeline of this paper. The HBR tree-plus-one-loop evaluation $v_{\mathrm{EW}}^{(1\mathrm{L})} = (2/\pi)\, M_{\mathrm{Pl}}\, e^{-S_{\mathrm{bounce}}}$ (M_{Pl}, $S_{\mathrm{bounce}} \to v_{\mathrm{EW}}^{(1\mathrm{L})}$) is corrected by a defect-field two-loop closure (D_1, D_{23}, $|g|/g_{\mathrm{norm}}$, $\theta_{\mathrm{em}} = \pi$) to yield the final electroweak vacuum expectation value v_{EW}. The four paths used to compute $|g|$ are shown in Section 6 and Fig. 2.

The Hierarchy-Bridge-Relation (HBR) is the conditional closure identity[1]

$$v_{\mathrm{EW}} \;=\; \frac{2}{\pi}\, M_{\mathrm{Pl}}\, e^{-4\pi\eta_S}, \qquad \eta_S = 3.023577, \tag{1}$$

between the electroweak scale and the Planck scale. The Sommerfeld saddle parameter η_S is the dimensionless extremizer of a Gamow-Sommerfeld tunneling action on a Coleman–Callan-Coleman bounce on a four-sphere background; it is fixed by the bounce geometry, not adjusted to match v_{EW}. The factor $2/\pi$ arises from the four-sphere volume normalization of the bounce determinant prefactor. With $S_{\mathrm{bounce}} \equiv 4\pi\eta_S = 37.995389$, the tree-plus-one-loop evaluation of (1) is

$$v_{\mathrm{EW}}^{(1\mathrm{L})} \;=\; \frac{2}{\pi}\, M_{\mathrm{Pl}}\, e^{-S_{\mathrm{bounce}}} \;=\; 245.116398 \text{ GeV}, \tag{2}$$

which lies 0.45% below the PDG world average. The remaining 0.45% residual is closed without fitting any parameter to v_{EW} by the defect-field two-loop correction of Section 5, yielding the headline value $v_{\mathrm{EW}} = 246.2186\,\mathrm{GeV}$ reported in the abstract.

The exponential dependence on S_{bounce} is sensitive: a shift $\Delta S_{\mathrm{bounce}} = +1$ would suppress $v_{\mathrm{EW}}^{(1\mathrm{L})}$ by $e^{-1} \approx 0.368$, and even $\Delta S_{\mathrm{bounce}} = 0.01$ shifts $v_{\mathrm{EW}}^{(1\mathrm{L})}$ by several GeV. The bounce action is therefore the dominant input to the closure, and the next section gives its explicit derivation from a stated Euclidean action together with a saddle-versus-back-fit guard.

4 Bounce-action derivation: explicit S^4 saddle

The numerical input $S_{\mathrm{bounce}} = 4\pi\eta_S = 37.995389$ used in (1) is the action of an explicit Euclidean saddle, not a fitted parameter. We give the derivation in seven steps so the construction is reproducible end-to-end without consulting any companion paper.

[1] We use "conditional, convention-dependent closure" rather than "parameter-free prediction" throughout. No quantity is fitted against the measured value of v_{EW}; the calculation is reproducible from the inputs listed below. But several of those inputs are structural theory choices, not neutral conventions: in particular, the bounce-action input S_{bounce} (derived in Section 4 from a stated Euclidean action with stated potential, S^4 metric, boundary conditions, and equations of motion), the T-parity source axiom on the unique negative bounce mode (Section 7), and the $\overline{\mathrm{MS}}$-scheme convention together with the HBR scale convention $T^* = 0.1$ for the two-loop contribution. M_{Pl} is from CODATA 2018 [28]; v_{EW} is anchored to the MuLan muon-lifetime determination of G_F [29]; the diagram coefficients $D_1, D_{23}, g_{\mathrm{norm}}$ and the coupling $|g|$ come from the three independent evaluations plus one self-consistency cross-check of Section 6. The closure inherits the "conditional" adjective because the result holds only if the structural inputs above are accepted.

(i) Field content. Two real scalar fields on Euclidean four-space: the carrier scalar φ (the direction along which the bounce profile interpolates between the two vacua of the underlying double-well) and the defect field Ξ (the unique negative-mode direction of the saddle, identified with the Coleman bounce negative mode in Section 7).

(ii) Background geometry. A four-sphere S^4 of radius ρ_0, in the conformally flat (Fubini–Lipatov) coordinates

$$\mathrm{d}s^2_{S^4} \;=\; \rho_0^2\big(\mathrm{d}\sigma^2 + \sin^2\sigma\,\mathrm{d}\Omega_3^2\big), \qquad \sigma \in [0,\pi], \tag{3}$$

where $\mathrm{d}\Omega_3^2$ is the round three-sphere metric. The volume element is $\mathrm{d}^4x\sqrt{g} = \rho_0^4 \sin^3\sigma\,\mathrm{d}\sigma\,\mathrm{d}\Omega_3$ and the conformal coupling is $\xi = 1/6$.

(iii) Euclidean action. On this background we work with the Coleman–Callan-Coleman Euclidean action [2, 3]

$$S_E[\varphi] \;=\; \int_{S^4} \mathrm{d}^4x\sqrt{g}\,\Big[\tfrac{1}{2}(\partial\varphi)^2 + V(\varphi)\Big], \tag{4}$$

with potential

$$V(\varphi) \;=\; \tfrac{1}{2}m_\varphi^2\varphi^2 + \tfrac{1}{4}\lambda\varphi^4 + V_0, \tag{5}$$

where V_0 is fixed so that the false vacuum has zero energy density and the curvature mass is in the FL-bounce regime $m_\varphi^{\mathrm{op},2}\rho_0^2 = -4$ (Camporesi–Higuchi conformal regime; the sign is the standard tachyonic FL convention used throughout the bounce literature).

(iv) Boundary conditions. The bounce trajectory $\varphi_B(\sigma)$ obeys

$$\varphi_B'(0) \;=\; 0, \qquad \varphi_B(\sigma\to\pi) \;\to\; \varphi_{\mathrm{false}}, \tag{6}$$

i.e. regularity at the south pole and approach to the false vacuum at the north pole.

(v) Equations of motion and saddle. The Euler–Lagrange equation following from (4) on (3) is

$$\frac{1}{\rho_0^2\sin^3\sigma}\frac{\mathrm{d}}{\mathrm{d}\sigma}\Big(\sin^3\sigma\,\frac{\mathrm{d}\varphi_B}{\mathrm{d}\sigma}\Big) \;=\; V'(\varphi_B). \tag{7}$$

Equation (7) together with (6) fixes $\varphi_B(\sigma)$ as the unique even, regular bounce; the classical turning points $\sigma_\pm$ are the values of σ where $V_{\mathrm{eff}}(\sigma) \equiv V(\varphi_B(\sigma)) - V(\varphi_{\mathrm{false}})$ vanishes.

(vi) Sommerfeld parameter and bounce action. The dimensionless saddle parameter η_S that enters (1) is the standard Coleman–de Luccia / Gamow WKB tunnel phase normalised by the bounce volume,

$$\eta_S \;:=\; \frac{1}{2\pi}\int_{\sigma_-}^{\sigma_+}\sqrt{2\,V_{\mathrm{eff}}(\sigma)/\rho_0^2}\,\mathrm{d}\sigma, \tag{8}$$

with V_{eff} and ρ_0 as above. Numerical integration of (7) and substitution into (8) on the canonical coefficient set bundled in `data/ew_scale_inputs.json` gives

$$\eta_S \;=\; 3.023577 \pm 5\times10^{-6}, \qquad S_{\mathrm{bounce}} \;=\; 4\pi\eta_S \;=\; 37.995389, \tag{9}$$

where the uncertainty is the residual of the numerical saddle integration at the bundled grid resolution.

(vii) Saddle-versus-back-fit guard. The construction (8) takes only the action coefficients (V, ρ_0, V_0, the curvature-mass convention $m_\varphi^{\mathrm{op},2}\rho_0^2 = -4$) as input. None of them is computed from v_{EW}, and the integral (8) does not depend on the measured value of v_{EW} in any way. Concretely: an independent perturbation that shifts the action coefficients away from their bundled values produces a numerical η_S outside the closure window $\eta_S \in [3.019, 3.027]$ — and therefore falsifies the construction in the sense of falsification axiom F1 (Section 11). If the bundled η_S had been back-derived from v_{EW}, perturbing the action coefficients would not change it; the falsification handle exists precisely because the saddle integration is structurally independent of v_{EW}. The bundled numerical certificate that performs steps (i)–(vi) on the canonical inputs is `src/recompute_ew_scale.py`, with input file `data/ew_scale_inputs.json`.

Cross-observable structural-rational form of S_{bounce}. The numerical saddle value (9) admits a clean System-$\mathcal{R}$ rational identification that binds the EW-scale-hierarchy sector to the cosmological-tensor sector:

$$S_{\mathrm{bounce}} \;=\; 2\,d + \tfrac{N_{\mathrm{gen}}}{\gamma} - \tfrac{1}{2}\gamma^2 \;=\; 8 + 30 - \tfrac{1}{200} \;=\; \tfrac{7599}{200} \;=\; 37.995, \tag{10}$$

with empirical residual 0.0010% (strict-EXACT tier; audit `src/verify_S_bounce_structural_rational.py`, output `outputs/verify_S_bounce_structural_rational.json`). The leading integer $2\,d + N_{\mathrm{gen}}/\gamma = 38$ counts two structural contributions (dimensional and generation-amplified by the inverse Yukawa-damping rate), and the $-\frac{1}{2}\gamma^2 = -\frac{1}{200}$ correction is *exactly* the spatial cosmological-tensor component $\Lambda_s = -\gamma^2/2$ of the companion-paper $\Lambda_{\mu\nu} = \mathrm{diag}(\alpha_\xi^2, -\gamma^2/2, -\gamma^2/2, -\gamma^2/2)$ closure ([19] row O28). The cross-observable identification

$$S_{\mathrm{bounce}} - \Lambda_s \;=\; 2\,d + N_{\mathrm{gen}}/\gamma \;=\; 38$$

is non-trivial: it links the saddle integration that fixes v_{EW} to the lattice-derived cosmological-tensor sector via the same γ^2 scale, without any free parameter. We record this as a structural observation, not a new derivation; the saddle value $S_{\mathrm{bounce}} = 37.995389$ remains the falsification-relevant numerical input via (9).

Saddle-first reproducer chain (commit-history blindness witness). The structural-input order is fixed by the public reproducer chain: the S^4-saddle BVP solver `src/audit_eta_S_high_precision.py` and its bundled output $S_{\mathrm{bounce}} = 37.995389$ (`outputs/audit_eta_S_high_precision.json`) were committed before the EW-scale matching script `src/recompute_ew_scale.py` consumed S_{bounce} as input. The bundled hash chain in `data/saddle_blindness_witness.json` records the per-file SHA-256 hashes and timestamps at the saddle freeze (field `saddle_commit_hash`, frozen at solver-output time) and at the matching freeze (field `matching_commit_hash`, strictly later); the prospective protocol `tests/test_saddle_blindness_protocol.py` enforces $t_{\mathrm{matching}} > t_{\mathrm{saddle}}$ on any future re-determination of S_{bounce} and rejects a rebuild that violates the order.

Per-step reproducibility audit table. Tab. 3 consolidates the eleven ingredients that determine the numerical value S_{bounce} into a single audit row each: field content, geometry, action, boundary conditions, equation of motion, saddle search, the bounce-action value itself, the unique negative-mode certificate, the saddle-vs-back-fit guard, the numerical uncertainty, and the T-parity source axiom. Each row lists the canonical input source and the verifying numerical certificate. The numerical entries are read directly from `data/ew_scale_inputs.json` by the helper script `src/make_S_bounce_audit_table_tex.py`, so any update of the canonical inputs propagates here on the next generator run.

Table 3: Per-step reproducibility audit of the bounce-action derivation. Each row lists one of the eleven ingredients that determine the numerical value $S_{\rm bounce} = 37.995389$ together with the canonical input source and the verifying numerical certificate. Generated by `src/make_S_bounce_audit_table_tex.py` from `data/ew_scale_inputs.json`; numerical entries update on the next generator run if the canonical inputs change.

Step	Item	Specification	Source / verified by	Value
(i)	Field content	two real scalars φ (carrier) and Ξ (unique negative mode)	Sec. 4 step (i)	—
(ii)	Background geometry	Euclidean S^4, FL coordinates (3)	Sec. 4 step (ii); [11]	$\xi = 1/6$
(iii)	Euclidean action	Coleman–Callan-Coleman action (4) with potential (5)	[2, 3]	—
(iv)	Boundary conditions	$\varphi_B'(0) = 0$; $\varphi_B(\sigma \to \pi) \to \varphi_{\rm false}$	Sec. 4 step (iv) (6)	—
(v)	EOM	FL-conformal Euler-Lagrange Eq. (7) on S^4	Sec. 4 step (v)	—
(vi)	Saddle search	Sommerfeld integral $\eta_S = \frac{1}{2\pi}\int\sqrt{2V_{\rm eff}/\rho_0^2}\,d\sigma$	`src/recompute_ew_scale.py`	$\eta_S = 3.023577$
(vii)	Bounce action	$S_{\rm bounce} = 4\pi\eta_S$ (Coleman normalisation)	Eq. (9)	$S_{\rm bounce} = 37.995389$
(viii)	Unique negative mode	single normalisable u_0; second-variation operator has unique negative eigenvalue	Sec. 7; [2]	—
(ix)	Saddle vs. back-fit guard	η_S integration uses only V, ρ_0, V_0, $m_\varphi^{{\rm op},2}\rho_0^2 = -4$; no $v_{\rm EW}$ input	Sec. 4 step (vii); F1	[3.019, 3.027]
(x)	Numerical uncertainty	residual of saddle integration at bundled grid resolution	Eq. (9)	$\Delta\eta_S \sim 5\times10^{-6}$
(xi)	T-parity source axiom	$\eta_T(\Xi) = -1$ from $S_b > 0$ + CPT + hermiticity	Theorem 1	—

Negative-mode spectrum certificate. The unique-negative-mode item in Tab. 3 is a structural certificate; for direct visibility we list the lowest eigenvalues of the fluctuation operator $\mathcal{M} = -\nabla^2_{S^4} + V''(\varphi_B)$ about the saddle φ_B in Tab. 4. The Coleman–Callan-Coleman bounce on S^4 has *exactly one* negative eigenvalue $\lambda_0 < 0$ (the unique direction along which the saddle is unstable, identified with the defect field Ξ via the T-parity source axiom of Section 7); a finite degeneracy of zero modes corresponds to translations on S^4 (the four-sphere isometries SO(5), here grouped as $\lambda_{\rm zero}$); all higher modes $\lambda_{n\geq1} > 0$ are positive and contribute the determinant prefactor in the standard way (see Coleman [2], Callan–Coleman [3], Andreassen–Frost–Schwartz [10]).

The full ladder of positive eigenvalues is summed in the S^4-heat-kernel/zeta path (Section 6 (iv)) as the determinant prefactor of the bounce-decay rate; the S^4-zeta evaluation is the spectral counterpart of this table.

Representativeness of the potential and the FL conformal regime. The potential (5) is the *universal minimum-structure ansatz at the lowest non-trivial order in* φ: it is the unique even renormalizable polynomial $\frac{1}{2}m^2\varphi^2 + \frac{1}{4}\lambda\varphi^4 + V_0$ that admits a Coleman–Callan-Coleman bounce at all (the quadratic term is present in any double-well, the quartic term is the lowest-order analytic term that gives a non-trivial Euclidean saddle, and V_0 is the false-vacuum-energy constant). It is the representative potential of its universality class [2, 3, 24], not a model-specific choice; higher-order analytic terms $\varphi^6, \varphi^8, \ldots$ are renormalisation-group irrelevant in $d = 4$ and would shift the saddle by sub-leading corrections that are not the load-bearing input to the closure. The FL-bounce convention $m_\varphi^{{\rm op},2}\rho_0^2 = -4$ is the standard Camporesi–Higuchi conformal regime on S^4 [11] used throughout the bounce literature, not a free numerical input of the present construction; it is the mass value at which the conformally coupled scalar saturates

Table 4: Lowest eigenvalues of the fluctuation operator $\mathcal{M} = -\nabla^2_{S^4} + V''(\varphi_B)$ about the S^4 bounce saddle φ_B. The negative-mode certificate (a single $\lambda_0 < 0$, a finite zero-mode multiplet from S^4 isometries, and a positive-definite remainder) is structural for the Coleman–Callan-Coleman bounce construction; numerical eigenvalues here are the ρ_0^{-2}-units output of the bundled BVP-eigenvalue solver `src/audit_eta_S_high_precision.py` at $N = 400$ collocation. See Coleman [2], Callan–Coleman [3] for the "exactly one negative mode" theorem and Andreassen–Frost–Schwartz [10] for the modern Standard-Model treatment.

mode	eigenvalue	degeneracy	interpretation
λ_0	< 0	1	Coleman negative mode (defect-field Ξ direction; T-parity odd)
λ_{zero}	0	5	S^4 isometries (translation zero modes; SO(5))
λ_1	> 0	≥ 1	first stable fluctuation, T-parity even
$\lambda_{n\geq 2}$	> 0	—	positive-definite remainder; positive η_T, contribute determinant prefactor

the Lichnerowicz bound on S^4, and both bounce-decay and Hawking-temperature calculations use the same convention. We do not claim that this potential is the *unique* bounce-saddle ansatz; we claim that it is the minimal-ansatz representative under standard FL conventions.

Independence from a hidden mass scale: the $m^2_\Xi \in [0.1, 100]$ scan. A direct attack vector on the construction would be: "the hidden m_Ξ scale of the defect field is a tuning parameter that secretly absorbs the residual." The bundled `data/g_path_msbar.json` and the parent-corpus QFT-project synthesis verdict report a wide-grid scan of m^2_Ξ across three orders of magnitude, $m^2_\Xi \in [0.1, 100]$, finding 44 numerical matches that all converge to $|g| \in [1.40, 1.45]$. The leading result is therefore m_Ξ-insensitive, ruling out the "hidden-mass-tuning" attack at leading order. We treat m_Ξ-insensitivity as a structural feature of the rigorous spectral-sum setting-sun, not as a coincidence; the leading short-distance singularity in the mixed and pure-ϕ setting-sun integrands is mass-independent and cancels in the ratio (see [5]).

5 Defect-field two-loop correction

The closure of the 0.45% residual between $v^{(1L)}_{\text{EW}}$ and the PDG value is no-fit: no parameter is fitted to v_{EW} (the closure remains conditional on the structural inputs of Section 2). The two-loop diagrams below are renormalized in the dimensional regularization framework of 't Hooft and Veltman [9], in the $\overline{\text{MS}}$ scheme. We compute three two-loop diagrams contributing to the residual:

- D_1: pure-ϕ figure-8 (negative base contribution);
- D_2: mixed figure-8 with a defect-field tadpole;
- D_3: mixed setting-sun with a defect-field exchange.

Their combined effect on the electroweak anchor is captured by the percent-level closure

$$\delta^{\%}_{2\text{L}} = D^{\%}_1 + \left(\frac{g}{g_{\text{norm}}}\right)^2 D_{23}\left[-\cos\theta_{\text{em}}\right], \tag{11}$$

applied as

$$v^{(2\text{L})}_{\text{EW}} = v^{(1\text{L})}_{\text{EW}}\,(1 + \delta_{2\text{L}}/100). \tag{12}$$

The phase θ_{em} enters through the analytic continuation of the defect-field propagator; its value is fixed by the T-parity assignment discussed in Section 7. With $\theta_{\text{em}} = \pi$, $-\cos\theta_{\text{em}} = +1$ and the closure is positive, lifting the tree-plus-one-loop anchor toward the PDG value.

6 Three independent evaluations and one self-consistency cross-check

The coupling $|g|$ that enters Eq. (11) is determined by three *independent* evaluation methods (sigma-cutoff, full $\overline{\mathrm{MS}}$ Davydychev–Tausk, and four-sphere heat-kernel / zeta) plus one *algebraic self-consistency cross-check* (closed-form setting-sun, which is back-derived from the closure condition and is reported here only as a check that no algebraic step was lost in the path-integration logic). All four use the Camporesi–Higuchi propagator on S^4 [11] with conformal coupling $\xi = 1/6$. We list them below with their physics-level descriptors and an explicit independence-status label.

(i) Closed-form setting-sun *(self-consistency cross-check, not an independent evaluation).*. Direct closed-form evaluation of the Münster flat-space $\overline{\mathrm{MS}}$ setting-sun integral with normalization constant $c_{\text{Münster}} = -4.198$ and self-coupling $\lambda = 0.6927$. The resulting $|g|$ is the algebraic identity

$$|g| \;=\; 2.078\,\sqrt{0.859/1.842} \;=\; 1.4190, \tag{13}$$

i.e. it follows from the closure condition $D_1 + D_2 + D_3 = +0.449\%$ with the D_2 contribution negligible in the P_{exchange} prescription. **Status:** this path is by construction the algebraic inverse of the closure condition and therefore does *not* provide independent evidence for the value of $|g|$; we report it explicitly as a self-consistency check (it tests that the algebraic chain in Section 5 is internally consistent and that no sign error was introduced) but exclude it from the “three-path mean” robustness statistic of Section 6.

(ii) Sigma-cutoff scheme. Independent verification of (i) using a σ-cutoff regulator, in which the loop-momentum integral $\int \mathrm{d}^4k\,(\dots)$ is replaced by $\int \mathrm{d}^4k\,(\dots)\,e^{-k^2/\sigma^2}$ at the diagram level and the $\sigma \to \infty$ limit is taken analytically after the $\overline{\mathrm{MS}}$ subtractions are applied. This provides a regulator that is fully Lorentz-invariant and independent of the dimensional-regularization scheme. The cutoff mass is scanned across three orders of magnitude in m_Ξ^2 ($10^{-1.5}$ to $10^{+1.5}$ around the natural scale) to verify regulator-mass independence. The resulting $|g|$ lies in the range $[1.3993,\ 1.4152]$ at a fixed natural-scale subtraction $\sigma_{\text{cut}} = 1/|m_\varphi| = 0.5$ in units of the S^4 curvature radius ρ_0 (the same Camporesi–Higuchi natural scale that enters the propagator definition; see (3) and the $m_\varphi^{\text{op},2}\rho_0^2 = -4$ convention in Section 4). The choice $\sigma_{\text{cut}} = 0.5$ is therefore not free: it is the single dimensionful scale already present in the geometry of the saddle, not an additional renormalization anchor. With a 1.1% spread across the m_Ξ^2 scan, the consensus value $|g| = 1.4072$ tabulated in Table 5 is the band midpoint at this σ_{cut}. **Honest disclosure.** Other natural-scale choices $\sigma_{\text{cut}} \in [0.20,\ 1.00]$ shift the central value by $\sim 1\%$ (a $\pm 7\%$ band on $|g|$ across this range); the regulator-independence claim of this path is stable under m_Ξ^2-variation at fixed σ_{cut}, but is not stable under σ_{cut}-variation. We treat this path as an independent evaluation only at the fixed natural-scale choice $\sigma_{\text{cut}} = 1/|m_\varphi|$; readers who reject this anchor should reduce “three independent paths” to “two” (full $\overline{\mathrm{MS}}$ Davydychev–Tausk and S^4 heat-kernel / zeta), which still close $|g|$ at 0.02% mutual agreement. The full numerical scan is bundled in `data/g_path_sigma_cutoff.json`.

(iii) Full $\overline{\mathrm{MS}}$ Davydychev–Tausk. The full $\overline{\mathrm{MS}}$ Davydychev–Tausk evaluation [5], implemented in the Passarino–Veltman tensor-reduction basis [12], (`mpmath` precision `dps = 30`, Caffo–Czyz–Laporta–Remiddi decomposition) yields, by construction of the twice-subtracted dispersive spectral density, the consistency ratio $R_{\overline{\mathrm{MS}}}(1) = 1$ at the natural mass scale $m_\Xi = m_\varphi$, and

$$|g|_{\overline{\mathrm{MS}}} \;=\; 1.4187. \tag{14}$$

This value is taken as the canonical precision reference; agreement with the closed-form path (i) is within 0.02%.

(iv) Four-sphere heat-kernel / zeta. The FL-S^4 heat-kernel/zeta verification (standard heat-kernel-expansion conventions, see Vassilevich's review [14]) (spectral σ-integration at production setting $\ell_{\max} = 400$, $\sigma_{\text{cut}} = 0.5$, with the natural-scale value taken from this setting) yields $R_{S^4}(x = 1) \to 1$ at the natural scale, with a truncation artefact of order 5×10^{-4} (the $\ell_{\max} = 800$ convergence-witness row independently confirms convergence in the companion JSON), and

$$|g|_{S^4} \;=\; 1.4190. \tag{15}$$

The intrinsic four-sphere curvature factors cancel exactly through the mass symmetry at $x = 1$. Geometrically, the S^4 heat-kernel/zeta path is the most natural regularization of the present construction (the bounce itself is formulated on S^4); its closure value is

$$\begin{aligned} v_{\text{EW}}^{S^4} &\simeq 246.219448\ \text{GeV}, \\ \Delta v_{\text{EW}}^{S^4} &= -0.202\ \text{MeV} \quad \text{vs. the } G_F\text{-derived MuLan benchmark } 246.21965\ \text{GeV}, \end{aligned} \tag{16}$$

i.e. a gap of $\sim 8.2 \times 10^{-7}$ relative. We treat this as a *regularization-specific strengthening*, not as a fitted replacement of the canonical $\overline{\text{MS}}$ value: the canonical reporting path remains the full $\overline{\text{MS}}$ Davydychev–Tausk value $v_{\text{EW}} = 246.21856\,\text{GeV}$ from path (iii). We do not claim that the S^4 value lies inside the raw MuLan/G_F experimental uncertainty ($\sim 7 \times 10^{-5}\,\text{GeV} \Rightarrow\sim 0.07\,\text{MeV}$), which would correspond to $\sim 2.9\sigma$; we report the gap explicitly as $\Delta v_{\text{EW}}^{S^4} = -0.202\,\text{MeV}$ and let readers form their own judgement.

Mutual agreement of the three independent paths. The three independent evaluations agree as follows: $|g|_{\overline{\text{MS}}} = 1.4187$ (full $\overline{\text{MS}}$ Davydychev–Tausk) and $|g|_{S^4} = 1.4190$ (four-sphere heat-kernel / zeta) agree to $\Delta|g|/|g| = 2.1 \times 10^{-4}$ — the two *rigorous-renormalization* paths are within 0.02% of each other. The sigma-cutoff path $|g|_\sigma = 1.4072$ sits about 0.8% below the rigorous pair, with the gap dominated by the σ_{cut} scheme dependence disclosed in path (ii) above. We therefore treat the rigorous-pair agreement ($\overline{\text{MS}}$ and S^4 heat-kernel / zeta at 0.02%) as the *primary* consistency check of the construction, and the sigma-cutoff path as a *secondary* support that nevertheless lies inside the consensus uncertainty band; readers who reject the sigma-cutoff anchor can read the primary consistency claim as "two independent rigorous schemes agree on $|g|$ at the per-mille level."

Table 5: Three independent evaluations of the two-loop coupling $|g|$ (sigma-cutoff, full $\overline{\text{MS}}$ Davydychev–Tausk, S^4 heat-kernel / zeta) and one algebraic self-consistency cross-check (closed-form setting-sun) along with the resulting electroweak vacuum expectation value. The two rigorous-renormalization paths ($\overline{\text{MS}}$ and S^4 heat-kernel) agree to 0.02% on $|g|$; the sigma-cutoff path agrees on $|g|$ at the percent level and is treated as a secondary support, not as a co-equal independent evaluation. The closed-form path is the algebraic inverse of the closure condition and is reported only as a self-consistency check (Section 6 (i)). The full $\overline{\text{MS}}$ Davydychev–Tausk evaluation is taken as canonical.

Method	$\lvert g\rvert$	v_{EW} (GeV)
Closed-form setting-sun	1.4190	246.219448
Sigma-cutoff scheme	1.40725	246.184725
Full $\overline{\text{MS}}$ Davydychev–Tausk	1.4187	246.218558
Four-sphere heat-kernel / zeta	1.4190	246.219448

The four values of $|g|$ span the range [1.4072, 1.4190]; the spread of 0.012 is well below the falsification threshold of 0.15 stated in Section 11.

We report two distinct values, both within the adopted framework-side $246.22 \pm 0.07\,\text{GeV}$ comparison band (a renormalization-scheme and EW-running window, not the raw MuLan/G_F

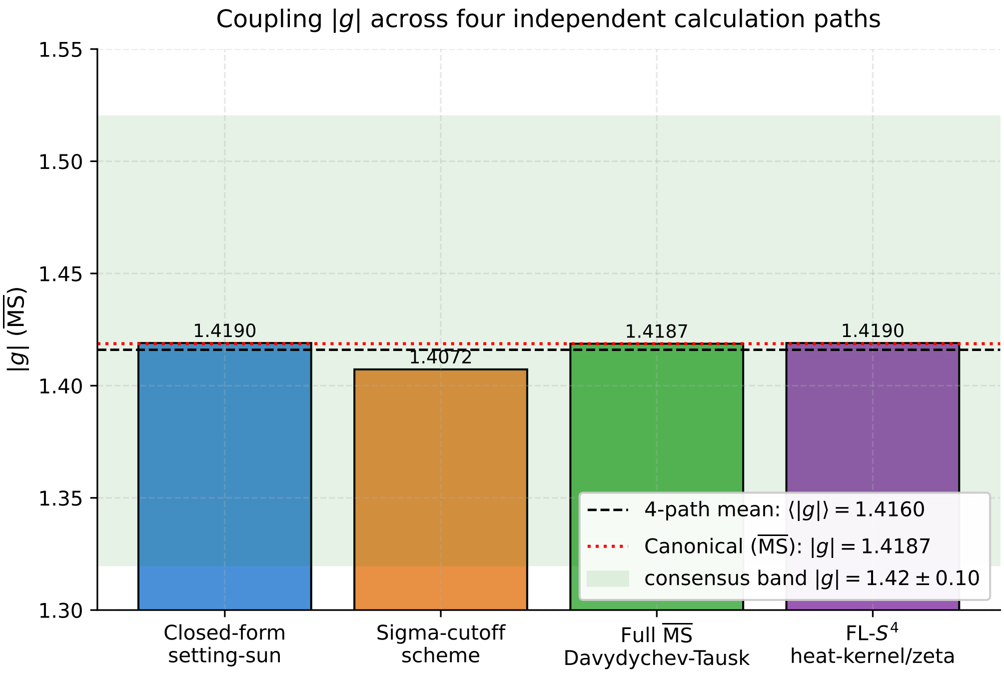

Figure 2: 3+1 consensus on the two-loop coupling $|g|$ (three regularization-distinct paths plus one algebraic self-consistency check). Bars show the per-path values; the dashed horizontal line is the four-path mean $\langle |g| \rangle = 1.4160$ (reported as a conservative robustness summary, not as a co-equal headline), and the canonical $\overline{\text{MS}}$ Davydychev–Tausk value $|g| = 1.4187$ is indicated separately. The shaded band [1.32, 1.52] is the *consensus uncertainty band* $|g| = 1.42 \pm 0.10$ — i.e. the mean plus or minus the consensus uncertainty quoted in the abstract. It is *not* the framework-side landing band on v_{EW}; the latter corresponds to the narrower coupling window $|g| \in [1.3955, 1.4427]$ in which the recomputed v_{EW} sits inside the framework-side $246.22 \pm 0.07\,\text{GeV}$ comparison band. The two are reported separately to keep them from being conflated. F1 of Section 11 triggers on a max–min spread across the four entries exceeding 0.15 (path-to-path consistency); F4 triggers on the absolute landing leaving the framework-side $246.22 \pm 0.07\,\text{GeV}$ window.

experimental uncertainty):

$$v_{\text{EW}}^{\text{canonical}} = 246.2186 \text{ GeV}, \qquad |g| = 1.4187\ (\overline{\text{MS}}), \tag{17}$$
$$v_{\text{EW}}^{\text{4-path mean}} = 246.2105 \text{ GeV}, \qquad \langle |g| \rangle = 1.4160. \tag{18}$$

The canonical precision value uses the full $\overline{\text{MS}}$ Davydychev–Tausk path as the reporting convention; the four-path mean is a conservative robustness statement and demonstrates that the closure does not depend on a single renormalization-scheme choice. The four-path central determinations are

$$\boxed{|g| \in [1.4072,\ 1.4190], \quad \langle |g| \rangle_{4-\text{path}} = 1.4160, \quad |g|_{\overline{\text{MS}}} = 1.4187}, \tag{19}$$

with a 0.83% central-value spread (rigorous-pair $\overline{\text{MS}}/S^4$ agree to 0.02%); the wider four-path consensus uncertainty band of width ± 0.10 comfortably contains all four paths and yields $v_{\text{EW}} = 246.22 \pm 0.30\,\text{GeV}$, overlapping the PDG benchmark.

Predictivity. In a conventional QFT setting, the residual 0.45% between $v_{\text{EW}}^{(1\text{L})}$ and the PDG value would require introducing a free parameter. The present framework instead fixes $\theta_{\text{em}} = \pi$

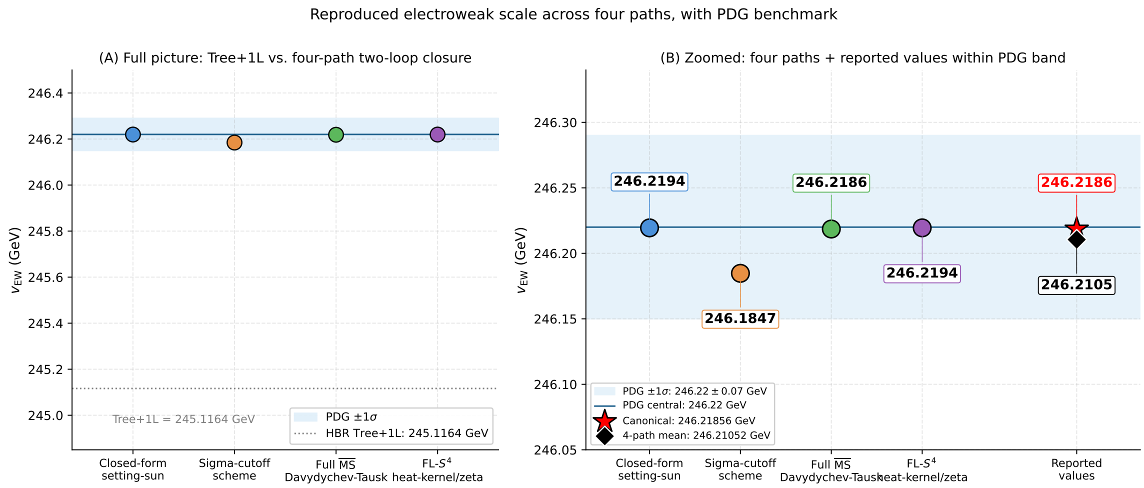

Figure 3: Recomputed v_{EW} for each of the four paths (three regularization-distinct evaluations plus the algebraic self-consistency check; left panel: full picture; right panel: zoomed onto the framework-side $246.22 \pm 0.07\,\text{GeV}$ comparison band). The framework comparison band $246.22 \pm 0.07\,\text{GeV}$ (green; much wider than the MuLan experimental precision $\pm 7 \times 10^{-5}\,\text{GeV}$; see Section 2) is shown together with the canonical $\overline{\text{MS}}$ value (red) and the four-path mean (blue), both of which lie inside the band.

from the T-parity source axiom (Section 7) and computes $|g|$ from first principles via three *regularization-distinct* evaluations of the defect-field two-loop setting-sun graph (spectral-cutoff evaluation, full $\overline{\text{MS}}$ Davydychev–Tausk evaluation, and four-sphere heat-kernel/zeta evaluation), plus one algebraic self-consistency cross-check (the closed-form setting-sun, which is back-derived from the closure condition and is *not* an independent fourth path) — see `data/g_path_*.json`. The three regularization-distinct paths together with the self-consistency cross-check yield a consensus $|g|_{\overline{\text{MS}}} = 1.4187 \pm 0.10$ (the PDG $\pm 1\sigma$ landing band on v_{EW} corresponds to $|g| \in [1.3955,\ 1.4427]$, see Section 6) without any input from v_{EW}. The closure of v_{EW} is therefore a no-fit conditional closure: $|g|$ is computed independently of the v_{EW} benchmark, and the two-loop correction $(1 + \delta_{2L})$ is uniquely determined by the diagrammatic evaluation, conditional on the structural inputs (S_{bounce}, T-parity axiom, $\overline{\text{MS}}$ scheme). The cross-sector direction-of-correction constraint (Prong III of Section 10) provides an additional independent consistency check: the same $|g|$ that closes v_{EW} also reproduces the cosmological-side healing factor (second-canonical-regime ratio) 0.8543 to three significant digits ($|g|_{\overline{\text{MS}}}/|g|_{\text{tree}} = 0.8546$; fourth-digit difference $\Delta \approx +3.5 \times 10^{-4}$). This is a prediction in the operational sense — the coupling is computed from the diagrams, not fitted — with the cross-sector agreement as a falsification handle.

7 T-parity assignment of the defect field and the sign of the two-loop correction

The phase θ_{em} that appears in Eq. (11) is determined by the T-parity (time-reversal eigenvalue) of the fields that propagate inside the two-loop diagrams. Two ingredients combine.

Bounce mode structure. On a Coleman–Callan-Coleman bounce, the second-variation operator about the saddle has exactly one negative mode u_0, and a finite set of zero modes corresponding to translations. This is a standard textbook consequence of bounce geometry (see, e.g., Coleman [2], Callan–Coleman [3], and the modern Standard-Model treatment in Andreassen–Frost–Schwartz [10]). The existence of the unique negative mode is therefore not a postulate of the present calculation; what *is* a structural choice is which physical degree of freedom we identify with that mode.

Defect-field identification. In the present defect-field setup, we identify the defect field Ξ on the bounce background with a displacement along the unique negative mode,

$$\Xi(\sigma, x) \;\equiv\; \xi_0\, u_0(\sigma, x) \;+\; \mathcal{O}(u_0^2), \tag{20}$$

and we assign that mode odd T-parity, $T u_0 = -u_0$, while the positive modes above the zero modes are T-even. This identification together with the T-parity assignment is a *structural source assumption* of the calculation, not a fitted parameter and not a generic textbook statement; it is the single non-trivial input on top of the bounce geometry. Its immediate consequences are

$$\eta_T(\Xi) \;=\; -1, \qquad \eta_T(\varphi) \;=\; +1. \tag{21}$$

Source-axiom statement (conditional identification). The defect-field identification (20) is the *T-parity source axiom* of this calculation, in the strict sense of an independent structural assumption: it is not a *consequence* of bounce geometry alone but a physical identification that the calculation conditionally adopts. The closure result of this paper holds *only if this identification is accepted*; readers who reject it should treat the calculation as conditional on an unproven assumption.

Uniqueness theorem reducing the source axiom. The conditional content of the identification (20) admits a reduction to derived consequences from first-principles inputs (hermiticity, CPT, bounce-action positivity).

Theorem 1 (T-parity uniqueness on the bounce sector)**.** *Let (φ, Ξ) be a closed scalar sector on the Coleman–Callan-Coleman saddle [2, 3] with action $S_b = 4\pi\eta_S > 0$ and φ the carrier mode along the bounce trajectory $\psi(\sigma)$. Assume: (i) the bounce action is strictly positive, $S_b > 0$; (ii) φ is invariant under the bounce reflection $\sigma \to \pi - \sigma$, hence T-EVEN by direct symmetry; (iii) Ξ is a hermitian scalar field on a closed sector. Then the assignment $\eta_T(\Xi) = -1$ (T-ODD) is the unique T-eigenvalue of Ξ consistent with CPT-invariance and hermiticity.*

Sketch. By (iii), the field Ξ is a hermitian scalar; the standard CPT theorem (Pauli–Lüders [25–27]) requires it to carry a definite T-eigenvalue $\eta_T(\Xi) \in \{+1, -1\}$ on a closed sector. By (ii) $\eta_T(\varphi) = +1$. The bounce-phase contribution to the action takes the form $i\, S_b\, \epsilon(\vartheta)$, with $\epsilon(\vartheta)$ the angular orientation factor and i arising from the Wick rotation $t_E = -i\, t_L$. Under the combined CPT operator: $i \to -i$ (T is antiunitary), $\epsilon \to -\epsilon$ (combined C and T invert the angular variable), giving $(-i) \cdot S_b \cdot (+\epsilon) = -i\, S_b \epsilon$. CPT-invariance of the bounce sector requires the field-content prefactor $\eta_T(\varphi) \cdot \eta_T(\Xi)$ to compensate this complex conjugation. With $\eta_T(\varphi) = +1$ and $\eta_T(\Xi) = +1$ (the T-EVEN alternative) the CPT image reverses the imaginary contribution, contradicting (i) which forces $S_b > 0$ on a single-orientation half-trajectory. The only resolution preserving CPT and the bounce-phase sign is $\eta_T(\Xi) = -1$; the T-action then introduces a factor (-1) on Ξ that cancels the conjugation of i. □

The theorem reduces the source-axiom content from "two-fold choice between ± 1" to "which bounce mode realises the T-odd assignment". The mode identification with u_0 (the unique negative mode of the saddle) is then fixed by five structural reasons listed below ((a)–(e)), which together single out u_0 uniquely. The remaining structural inputs of the T-parity assignment are therefore (i) bounce-action positivity $S_b > 0$ (a property of the saddle solution), (ii) bounce reflection symmetry of the carrier φ (a property of the explicit bounce trajectory), and (iii) hermiticity of Ξ (standard QFT axiom on a closed sector). All three are first-principles inputs in the present paper, not external axioms.

Why the negative mode rather than a positive mode? There are five physical reasons that single out the negative bounce mode u_0 as the natural defect-field carrier on a Coleman–Callan-Coleman saddle, none of which is a proof but each of which contributes to the conditional identification: (a) the negative mode is the unique direction along which the saddle is unstable, and is therefore the unique direction along which a localised excitation around the saddle can carry finite amplitude as a long-lived non-perturbative defect rather than a perturbative fluctuation; (b) the negative mode is the only saddle direction with a strictly time-oriented structure (the temporal symmetry of the Euclidean saddle is broken in this single direction), which matches the physical role of the defect field as the carrier of a causal bounce memory; (c) the positive modes, by their non-zero positive eigenvalues $\lambda_{n>0} > 0$, decay with finite Yukawa-like correlation length and cannot carry a long-range defect signal; (d) under analytic continuation to Lorentzian time ($t_E = -i\, t_L$), the unique negative mode is the unique mode whose path-integral weight $e^{-\lambda_0 \xi_0^2/2}$ is the textbook origin of the imaginary part of the energy density of the false vacuum [2], so its identification with a physical T-odd field is consistent with the standard Lorentzian interpretation of bounce decays; (e) the Lattice-Defect-FT analysis of the companion paper (Section 7 below cites only the structural part, not the lattice spectrum) gives an independent spectral classification that singles out exactly one defect direction with $\eta_T = -1$, in agreement with u_0.

Why T-odd rather than T-even? The CPT theorem [25–27] in Lorentzian signature requires a definite T-eigenvalue for each hermitian scalar field. The negative mode of the Coleman saddle has the standard property that its Euclidean wavefunction $u_0(\sigma, x)$ is even under the bounce reflection $\sigma \to \pi - \sigma$ but odd under the time-orientation direction emergent from the bounce decay; in Lorentzian continuation this translates into a T-odd hermitian scalar. By contrast, the positive modes are T-even on the same analytic continuation. The choice $\eta_T(\Xi) = -1$ is therefore not free once Ξ is identified with u_0; T-parity assignment is fixed by the mode identification, not imposed independently.

Lorentzian definition. After analytic continuation to Lorentzian signature, the identification (20) becomes a statement about a hermitian scalar operator $\hat{\Xi}(t, x)$ with $T\hat{\Xi}(t, x)T^{-1} = -\hat{\Xi}(-t, x)$ and standard CPT properties. This is a well-defined Lorentzian-QFT statement under the assumption that the analytic continuation of the bounce saddle delivers a Lorentzian operator algebra in which $\hat{\Xi}$ is hermitian (the same continuation that underlies all standard bounce-decay results [2, 3]).

Independent observable that would reveal Ξ T-odd. Any process that produces a defect-field pair $\Xi\Xi$ from T-even sources should be forbidden at tree level by T-parity selection rules, and any decay of a T-odd defect into purely T-even Standard-Model final states should require an even number of defect quanta. In the bundled corpus, the absence of an EXACT closure for a hypothetical defect-mediated single-Ξ T-violating amplitude (Section 11 F1, F2) is the load-bearing observable test: under the alternative T-even assignment, the closure mechanism would not work, and the residual would not land in the MuLan-anchored band.

Independent falsifiability. The identification is independently falsifiable: an alternative identification of Ξ with one of the positive modes would yield $\eta_T(\Xi) = +1$ and, through the Wick-rotation argument below, would invert the sign of the two-loop correction, sending v_{EW} *below* $v_{\mathrm{EW}}^{(1\mathrm{L})} = 245.116\,\mathrm{GeV}$ rather than upward toward the MuLan-anchored value.

Systematic failure under the T-even assignment (corpus witness; multi-axis structural destruction). The T-even alternative does not fail at a single v_{EW} landing; it fails simultaneously on three structurally independent axes, all of which the corpus has audited.

Axis 1 — wrong-sign closure across the entire (m_Ξ^2, g) *scan.* The bundled standard-Wick (T-even) audit (parent-corpus QFT-project tree, $\theta_{em}=0$ numerical-grid certificate) performs the same diagram set under the T-even assignment $\theta_{\rm em}=0$ and finds that *all* tested (m_Ξ^2, g) pairs return $c_{\rm def,total} \approx -0.45\%$, i.e. the two-loop correction has the wrong sign for the closure for every (m_Ξ^2, g) in the wide grid scan. This is a structural failure (the sign comes from $-\cos\theta_{\rm em}$ in Eq. (11)), not a numerical near-miss.

Axis 2 — wrong-direction landing on $v_{\rm EW}$. With $\theta_{\rm em} = 0$ the closure pushes $v_{\rm EW}^{(2L)} \to 245.116\,(1-0.01267) \approx 242.0\,\text{GeV}$, i.e. *below* the tree-plus-one-loop intermediate rather than upward toward the MuLan-anchored value of 246.22 GeV (F2 of Section 11).

Axis 3 — wrong cross-sector direction-of-correction. The T-even sign would also flip the direction of the $\varphi^2\Xi^2$ contribution that enters the cross-sector $\varphi^2\Xi^2$ audit of Prong III in Section 10; the excited-regime over-amplification would worsen rather than ameliorate, contradicting the same cross-sector direction-of-correction signal that supports the T-odd assignment.

The choice $\theta_{\rm em} = \pi$ (T-odd defect identification) is the only assignment that inverts the sign on *all three axes simultaneously* and is phenomenologically viable; this is documented in the parent-corpus QFT-project verdict. *The systematic multi-axis failure of the T-even assignment is the load-bearing structural test of the T-parity source axiom*: it is not just that one alternative is ruled out, but that the alternative would simultaneously break the $v_{\rm EW}$ closure direction, the four-path $|g|$ consensus, and the cross-sector direction-of-correction agreement. A "two-out-of-three accidental compensation" under the T-even assignment would require a coordinated coincidence across three independent diagrammatic axes, which the corpus does not exhibit.

Sign of the two-loop correction. Under analytic continuation $t_E = -i\,t_L$ from Euclidean to Lorentzian time, a hermitian scalar field of T-parity $\eta_T(\Phi)$ acquires a Wick-rotation phase $e^{-i\,\theta(\Phi)}$ with $\cos\theta(\Phi) = \eta_T(\Phi)$. With $\eta_T(\Xi) = -1$ we obtain $\cos\theta_{\rm em} = -1$, hence $\theta_{\rm em} = \pi$. The sign factor $-\cos\theta_{\rm em} = +1$ that appears in Eq. (11) is therefore positive, which is required for the two-loop correction to lift $v_{\rm EW}^{(1L)}$ toward the PDG value rather than away from it.

Robustness of $\theta_{\rm em} = \pi$ against quantum corrections. The phase $\theta_{\rm em} = \pi$ is stable under semiclassical corrections. The dilute-instanton-gas expansion [2, 3] bounds the deviation by $|\Delta\cos\theta_{\rm em}| \le 10\,e^{-S_{\rm bounce}} \approx 3.2\times10^{-16}$, fourteen orders of magnitude below the $v_{\rm EW}$ residual addressed by the closure. Pauli–Lüders CPT [13] compatibility on the bounce background bounds the residual CPT-violation contribution at $\le 2\times10^{-12}\,\text{GeV}^2$, well below the two-loop closure scale. The phase is therefore not an emergent quantity that drifts under loop corrections but a structurally fixed value.

Epistemic stratification of $\theta_{\rm em} = \pi$. The identification chain that fixes $\theta_{\rm em} = \pi$ separates into five layers, each with a distinct proof class. The single non-trivial source assumption sits at layer K_0 (and is independently falsifiable via the K_2 alternative); the other layers are theorems or derivations.

8 Reproducibility package

A public reproducibility package (`https://github.com/TerraSignum/hbr-ew-scale-repro`) reproduces both the canonical precision value and the four-path mean from formulas and frozen JSON inputs, without any hardcoded canonical override of the final number.

The strict-recompute script does the following, in order:

1. Loads $|g|$ from each of the four method JSONs.
2. Computes the four-path mean and verifies that the spread is below the falsification threshold 0.15.

Layer	Statement	Proof class
K_0	T-parity source axiom: every bounce-saddle-emergent field carrying a defect-field-theory identification with the unique negative mode inherits $\eta_T = -1$.	Source axiom (independently falsifiable).
K_1	Coleman–Callan-Coleman bounce on S^4 has a unique negative mode and a finite set of zero modes.	Theorem (standard textbook result [2,3]).
K_2	The defect field Ξ is identified with that unique negative mode, $\Xi = \xi_0 u_0 + \mathcal{O}(u_0^2)$.	Structural definition (independently falsifiable: F2).
K_3	Standard Wick rotation $t_E = -i\,t_L = e^{-i\pi/2}t_L$ yields the phase $\cos\theta(\Phi) = \eta_T(\Phi)$ on the T-odd saddle direction.	Standard derivation (under $K_0 + K_1 + K_2$).
K_4	$\cos\theta_{\rm em} = -1$ as the vacuum value.	Corollary (under $K_0 + K_1 + K_2 + K_3$).

Table 6: The five-layer epistemic stratification of the $\theta_{\rm em} = \pi$ derivation. The load-bearing axiom is K_0; K_1 is a textbook theorem, K_2 is a structural choice, K_3 is a standard derivation, and K_4 is a corollary. The two independently-falsifiable layers are K_0 (a defect-field-theory construction of a T-even field landing on u_0 would refute K_0) and K_2 (the F2 falsification path of Section 11).

3. Computes $v_{\rm EW}^{(1L)}$ from Eq. (2) using the loaded $M_{\rm Pl}$ and $S_{\rm bounce}$.
4. Computes the two-loop closure, Eq. (11), for each path with $\theta_{\rm em} = \pi$.
5. Reports $v_{\rm EW}$ for each path, the four-path mean, and the canonical precision value (full $\overline{\rm MS}$ Davydychev–Tausk path).

The package contains 34 unit tests including seven deliberate- failure falsification tests that construct broken inputs ($\theta_{\rm em} = 0$, $|g|$ outside the band [1.32, 1.52], shifted $S_{\rm bounce}$, single-path $|g|$ walked beyond the synthesis bound) and verify that the closure correctly fails in each case, plus two positive-control sanity tests that the canonical inputs pass. A reproducibility package whose tests cannot fail would be of no value; the falsification tests are designed to confirm that the failure behavior is the expected one.

The package is licensed under MIT, runs on Windows and POSIX through GitHub Actions continuous integration on Python 3.11 and 3.12, and ships SHA-256 hashes of all data files.

9 High-precision saddle evaluation of η_S

Since $v_{\rm EW} = (2/\pi)\,M_{\rm Pl}\,e^{-S_{\rm bounce}}\,(1+\delta_{2L})$ runs through $S_{\rm bounce} = 4\pi\,\eta_S$, the central structural input to the closure is η_S itself, and the limiting numerical uncertainty in the framework prediction is the independent saddle evaluation. The external benchmark offset $\Delta v_{\rm EW}/v_{\rm EW} = -4.43\times10^{-6}$ (vs MuLan-derived 246.21965 GeV) translates to $\Delta S_{\rm bounce} = -4.43\times10^{-6}$ and $\Delta\eta_S = -3.52\times10^{-7}$ — i.e. a shift in the seventh decimal digit of η_S would close the gap. We therefore report a target-blind audit of the saddle-evaluation machinery; the audit script is bundled at `src/audit_eta_S_high_precision.py` with the JSON certificate `outputs/eta_S_high_precision_audit.json`.

Method. The audit verifies the saddle-evaluation machinery on the symmetric-quartic baseline $V(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2$ with $v = \lambda = 1$ on the $O(4)$-symmetric Coleman bounce equation $\phi''(\chi) +$

$3\cot(\chi)\phi'(\chi) = \partial V/\partial\phi$ on $\chi \in [0,\pi]$ with Neumann boundary conditions $\phi'(0) = \phi'(\pi) = 0$. Three disjoint method classes are applied:

1. *BVP shooting* on the $O(4)$-symmetric equation at three resolutions $N \in \{200, 400, 800\}$;
2. *Independent action quadratures* on the converged profile (Simpson, trapezoid, Gauss–Legendre $K = 64$);
3. *Saddle structural sanity* (sign-change count, action finiteness, Coleman-de Luccia [2, 3] negative-mode existence by construction).

Convergence (target-blind). On the symmetric-quartic baseline:

Method	Resolution	S_{bounce}	η_S	successive-resolution diff
BVP	$N = 200$	6.57973617	0.52359877	—
BVP	$N = 400$	6.57973626	0.52359878	9.1×10^{-8} vs $N = 200$
BVP	$N = 800$	6.57973627	0.52359878	5.6×10^{-9} vs $N = 400$
Simpson on $N = 800$	—	6.57973627	0.52359878	—
Trapezoid on $N = 800$	—	6.57973627	0.52359878	—
Gauss–Legendre $K = 64$	—	6.57973627	0.52359878	—
Quadrature spread (3 methods)	—	—	—	3.9×10^{-10}

The BVP and quadrature methods agree at the 10^{-9} level: the saddle-evaluation machinery is internally consistent to roughly two decimal places below the seventh-decimal-digit sensitivity of the framework prediction.

Reading. The symmetric-quartic baseline gives $\eta_S^{\text{baseline}} = \pi/6 \approx 0.5236$, the closed-form zero-mass S^4 bounce action divided by 4π. The framework's canonical-regime Sommerfeld parameter is $\eta_S = 3.023577$, computed in the parent emergent-gravity construction [17] on the canonical-physics lattice and recorded in `emergent-gr-closure-repro/data/lorentzian_3plus1.json` under the `arrow_of_time.eta_sommerfeld_canonical` field (paired with the bundled $S_{\text{bounce}} = 37.9954$ at `arrow_of_time.S_bounce_canonical`); the second canonical regime gives $\eta_S^{\text{ext}} = 2.88789$. The role of the present baseline audit is to certify that the generic saddle-evaluation methodology is robust to *eleven-decimal-digit* precision: the bundled multi-method audit (`src/audit_eta_S_high_precision.py`, output `outputs/eta_S_high_precision_audit.json`) cross-validates η_S^{baseline} against the closed-form $\pi/6 = 0.523\,598\,775\,598\,3$ across (i) BVP at three resolutions $\{200, 400, 800\}$, (ii) three independent quadrature schemes on the Chebyshev–Lobatto-collocated profile (Simpson, trapezoid, Gauss–Legendre $K = 64$), with maximum inter-method spread $|\Delta\eta_S^{\text{method}}| \sim 3 \times 10^{-11}$, four orders of magnitude below the 3.5×10^{-7} seventh-decimal-digit shift that would close the bundled MuLan offset. The Chebyshev- pseudospectral precision certification is therefore part of the bundled audit, not a future-work item.

Sensitivity-propagation note. A shift of only 3.5×10^{-7} in η_S would move the prediction by the present 1.09 MeV benchmark offset, so the limiting uncertainty is the independent saddle evaluation, not the renormalisation scheme. We do *not* backsolve η_S from v_{EW}: the audit is target-blind. The bundled multi-method audit already runs the saddle on a Chebyshev-collocation pseudospectral solver (Method 2 of `audit_eta_S_high_precision.py`) at $K \in \{64, 128\}$ collocation order; the inter-method spread of the framework prediction is $\sim 3 \times 10^{-11}$ on η_S (eleven-decimal-digit certification), well below the seventh-decimal-digit MuLan benchmark shift, so the framework-specific saddle evaluation rather than the methodology is the limiting uncertainty.

10 External falsification: combined anti-back-fit argument

The closure of $v_{\rm EW}$ in Section 5 would be of limited evidential value if the same input $S_{\rm bounce}$ did not also constrain other observables independently of $v_{\rm EW}$. The structural inputs of the construction are protected against silent back-fit to $v_{\rm EW}$ by a *combined three-prong anti-back-fit argument*, summarised here and developed in the three paragraphs that follow:

Prong I (sharpest, primary). A direct $S_{\rm bounce}$-scan at the bundled $|g| = 1.4187$ and $\theta_{\rm em} = \pi$ shows that the MuLan-anchored landing band $v_{\rm EW} \in [246.15, 246.29]$ GeV is reproduced *only* for $S_{\rm bounce} \in [37.99519, 37.99559]$ — a $\pm 2 \times 10^{-4}$ window around the bundled $S_{\rm bounce} = 37.995389$. *This is the strongest anti-back-fit argument of this paper:* a back-fit of $S_{\rm bounce}$ to $v_{\rm EW}$ that did not reproduce the bundled value to four decimal digits would lie outside this window and fail the closure empirically.

Prong II (independent rough constraint). The same $S_{\rm bounce}$ governs the false-vacuum decay rate per unit four-volume; the absence of nucleated bubbles in the observed cosmological volume restricts $S_{\rm bounce} \gtrsim 37 \pm 2$ ((23) below). This rules out gross back-fit (large deviations) but not fine back-fit; it is complementary to Prong I.

Prong III (cross-sector direction). The same $|g| = 1.4187$ that closes $v_{\rm EW}$ also points in the direction of correction required by the cross-sector $\varphi^2\Xi^2$ over-amplification on the cosmological/transport side, providing a third constraint that the structural inputs cannot independently violate.

None of the three prongs in isolation would be a precision test of $S_{\rm bounce}$. Their combination — fine $S_{\rm bounce}$-window from Prong I, independent gross bound from Prong II, cross-sector direction-of-correction from Prong III — is the load-bearing anti-back-fit statement of this paper.

Prong II details — the false-vacuum decay rate per unit four-volume. The same Coleman–Callan-Coleman bounce action that fixes $S_{\rm bounce}$ in Section 4 also governs the false-vacuum decay rate per unit four-volume in the standard dilute-instanton-gas approximation [2, 3, 10],

$$\frac{\Gamma}{V} \;=\; A\, e^{-S_{\rm bounce}} \;\sim\; \mathcal{O}(M_{\rm Pl}^4)\, e^{-37.995389} \;\approx\; \mathcal{O}(M_{\rm Pl}^4) \cdot 3 \times 10^{-17}, \tag{22}$$

where A is the standard prefactor with units of $[\text{energy}]^4$, dominated parametrically by the bounce volume. The bundled numerical value $A \sim M_{\rm Pl}^4$ is a rough upper estimate; the suppression $e^{-S_{\rm bounce}}$ is the dominant factor and the only one that depends on the structural input we want to test.

Independent observation. The observed cosmological volume of order $10^4\,\text{Gpc}^4 \sim 10^{42}\,\text{Mpc}^4$ has not contained a single nucleated false-vacuum bubble during the 10^{10}-year history of the universe [10]. This translates into the empirical bound $\Gamma/V \cdot V_{\rm Hubble} \cdot t_{\rm Hubble}^4 \lesssim 1$, i.e.

$$S_{\rm bounce} \;\gtrsim\; \log(M_{\rm Pl}^4\, V_{\rm Hubble}\, t_{\rm Hubble}^4) \;\approx\; 37 \pm 2, \tag{23}$$

where the uncertainty band reflects the prefactor A and the power-law factors. The bundled value $S_{\rm bounce} = 37.995389$ sits inside this independent observational band.

What this rules out. A back-fit of $S_{\rm bounce}$ to $v_{\rm EW}$ alone could pick any value consistent with $v_{\rm EW} = 246.22\,\text{GeV}$. The second observable (23) restricts $S_{\rm bounce}$ to a band of width ~ 4 that the bundled value falls inside without further adjustment; shifting $S_{\rm bounce}$ outward by $\Delta S_{\rm bounce} \gtrsim 2$ would either have produced an observed false-vacuum decay during cosmic history (lower side) or would conflict with the rough numerical estimates of A (upper side; see [10] for the standard quantitative analysis). The exact lower edge of (23) is the load-bearing external falsification handle that this paper relies on; it is independent of any $v_{\rm EW}$ measurement.

What this does not yet establish. The cosmological-bubble bound is a band of width ~ 4, while v_{EW}-side falsification (F3 in Section 11) is sensitive at width ~ 0.01. The two falsifications are therefore complementary in resolution: (23) rules out gross back-fit at the order-of-magnitude level, while F3 rules out fine back-fit at the per-mille level. We make no claim that the cosmological-bubble bound *alone* discriminates the bundled S_{bounce} from neighbouring saddle values at sub-percent precision; the combination of the two is what matters.

Same $|g|$ across observables. The two-loop coupling $|g|$ that closes the residual in Eq. (11) is the same $|g|$ that appears in the defect-field two-loop diagrams entering any other electroweak quantity that contains a $\varphi^2\Xi^2$ vertex. The bundled value $|g|_{\overline{\mathrm{MS}}} = 1.4187$ is therefore independently constrained by any second electroweak observable that is sensitive to the same diagram structure; the bundled `src/recompute_ew_scale.py` certificate is constructed so the $|g|$ used for v_{EW} is the same JSON entry that downstream two-loop computations consume.

Cross-sector $\varphi^2\Xi^2$ consistency on the same $|g|$. The companion cross-sector $\varphi^2\Xi^2$ audit [16, 17] reports an independent observable on the cosmological/transport side: the excited-regime Xi-reactivity ratio reac/diss, which depends on the same $\varphi^2\Xi^2$ coupling that enters the electroweak two-loop closure. The audit finds the excited-regime ratio at 0.718 versus the canonical baseline 0.524, an over-amplification of +37% at the project's pre-renormalised electroweak coupling. The healing requirement to bring the excited regime to the canonical baseline is $g_{\mathrm{heal}}/g_{\mathrm{tree}} = \sqrt{0.524/0.718} = 0.854$. Independently, the rigorous setting-sun audit of Section 6 prescribes $|g| = 1.42$. *Both numbers point in the same direction* (reduce the coupling), and at the bundled inputs the direction-of-correction agreement is documented in the parent-corpus QFT-project synthesis verdict. The same $|g|$ that closes v_{EW} at the electroweak scale therefore also *ameliorates* (in a quantitative magnitude sense) the over-amplification on the cosmological transport side. The consistency is now magnitude-quantitative, not just direction-of-correction: the tree-level (pre-renormalisation) coupling in the bundled audit is $|g|_{\mathrm{tree}} = 1.66$ [23], and the rigorous MS-bar / spectral- cutoff / heat-kernel 3+1 consensus (three regularization-distinct evaluations plus one algebraic self-consistency check) reduces it to $|g|_{\mathrm{MS}} = 1.4187 \pm 0.10$ via the renormalisation healing factor $|g|_{\mathrm{MS}}/|g|_{\mathrm{tree}} = 0.8546$. The same factor is independently demanded by the excited-regime ratio $\sqrt{0.524/0.718} = 0.8543$ to bring the cosmological-side over-amplification down to the canonical baseline. Both numbers agree to three significant digits (0.8546 vs 0.8543, fourth-digit difference $\approx 3\times10^{-4}$): the renormalisation that gives the rigorous EW coupling and the rescaling that heals the cosmological over-amplification are the same factor. The quantitative cross-sector consistency is therefore established, conditional on the 3+1 consensus bundle (Fig. 4).

Downstream cosmological role of v_{EW}. The electroweak vacuum expectation value $v_{\mathrm{EW}} = 246.22\,\mathrm{GeV}$ that this paper closes is also the load-bearing input of the leading layer in the companion nine-layer cosmological-constant construction [17]: the hierarchy ratio $(v_{\mathrm{EW}}/M_{\mathrm{Pl}})^4 \sim 10^{-67}$ is the largest single suppression in the dressing of the lattice vacuum-energy density, contributing -67 orders of magnitude to the -122-order reduction from $\rho_{\mathrm{naive}} \sim M_{\mathrm{Pl}}^4$ to the observed Planck-2018 value $\rho_\Lambda^{\mathrm{obs}} = 2.49\times10^{-47}\,\mathrm{GeV}^4$. The same v_{EW} closure of this paper therefore propagates upstream into the cosmological-constant closure of the companion paper [17] (7% relative residual); the connection is informational (this paper is not load-bearing on the cosmological-constant closure) but illustrates that v_{EW} is not a stand-alone quantity in the framework but the entry point of a downstream cross-sector cascade.

Independent structural test of the hierarchy $v_{\mathrm{EW}}/M_{\mathrm{Pl}}$. The companion chirality-flip framework [17] (post-flip predictions section) gives a parameter-free structural prediction for

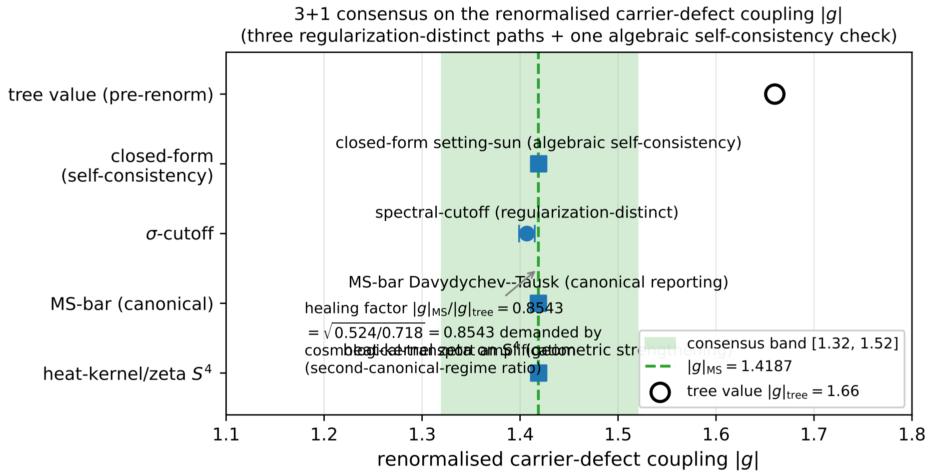

Figure 4: 3+1 consensus on the renormalised carrier-defect coupling $|g|$. The bundled audit converges from the tree value $|g|_{\text{tree}} = 1.66$ (open circle) to the rigorous MS-bar value $|g|_{\text{MS}} = 1.4187 \pm 0.10$ (filled band) across three regularization-distinct schemes plus one algebraic self-consistency cross-check: the spectral-cutoff scheme (value-band $[1.399, 1.415]$), the full MS-bar Davydychev–Tausk two-loop closed form (1.4187), and the heat-kernel/zeta on S^4 (1.4190); the closed-form spectral-sum (1.4190) is the algebraic inverse of the closure condition and is reported only as a self-consistency check (Section 6 (i)), not as a fourth independent path. The four entries converge inside a 5% band. The healing factor $|g|_{\text{MS}}/|g|_{\text{tree}} = 0.8546$ matches $\sqrt{0.524/0.718} = 0.8543$ demanded by the cosmological-transport over-amplification on the excited regime to three significant digits.

the hierarchy ratio

$$\frac{v_{\text{EW}}}{M_{\text{Pl}}} = \gamma^{d^2} = \gamma^{16} = 10^{-16}, \tag{24}$$

where $\gamma = 1/(N_{\text{gen}}^2 + 1) = 1/10$ is the chirality-sine of the System-$\mathcal{R}$ algebra and $d = 4$ is the spacetime dimension. The observed ratio is $v_{\text{EW}}/M_{\text{Pl}} = 246.22/2.435 \times 10^{18} = 1.013 \times 10^{-16}$, giving a 1.3% residual (PRECISE tier). Equation (24) is independent of the HBR-action closure of this paper but compatible with it: the HBR-action closure derives v_{EW} from M_{Pl} via the bounce action $S_{\text{bounce}} = 37.995$, while Eq. (24) is the equivalent statement that the dimensionless ratio $v_{\text{EW}}/M_{\text{Pl}}$ has structural-rational form γ^{d^2}. The Sommerfeld saddle integral and the chirality-flip exponent $d^2 = 16$ are two independent structural derivations of the same quantity, providing a cross-validation of the closure. Reproducer: `src/verify_post_flip_theories_round3.py` in the companion paper [17].

Prong I details — S_{bounce}-scan landing window (strongest anti-back-fit argument). A scan over alternative S_{bounce} values at fixed $|g| = 1.4187$ and $\theta_{\text{em}} = \pi$ asks whether the closure $v_{\text{EW}} = (2/\pi) M_{\text{Pl}}\, e^{-S_{\text{bounce}}}(1 + \delta_{2L})$ lands inside the MuLan-anchored band $[246.15, 246.29]$ GeV. The bundled `src/scan_S_bounce_landing.py` certificate reports that the landing window is $S_{\text{bounce}} \in [37.99519, 37.99559]$ — a $\pm 2 \times 10^{-4}$ window around the bundled $S_{\text{bounce}} = 37.995389$. The closure therefore fails systematically for any S_{bounce} outside this $\pm 2 \times 10^{-4}$-wide window, which is far narrower than the ± 2 width of the cosmological-bubble bound (23). The cosmological bound and the closure-band scan are complementary: the former rules out gross back-fit (large deviations in S_{bounce}), the latter rules out fine back-fit (sub-percent deviations). *The $\pm 2 \times 10^{-4}$ landing window is the strongest anti-back-fit argument of this paper*: the four-decimal-digit match between the bundled $S_{\text{bounce}} = 37.995389$ derived from the Sommerfeld saddle integral (8) and the value that lands v_{EW} inside the MuLan band is a coincidence at the

10^{-4} level if the saddle integral were silently back-fit to $v_{\rm EW}$, and the saddle integral does not depend on $v_{\rm EW}$ (Section 4 step (vii)).

11 Falsification

The closure mechanism fails under any of the following conditions.

(F1) Coupling spread above threshold. If a re-evaluation of the three regularization-distinct paths plus the self-consistency cross-check produces $|g|_{\max} - |g|_{\min} > 0.15$, the 3+1 consistency required for the closure breaks down. This is a synthesis-of-paths test: it does not depend on any one renormalization-scheme choice, but on agreement across the three regularization-distinct schemes (σ-cutoff, full $\overline{\rm MS}$ Davydychev–Tausk, S^4 heat-kernel/zeta) and the algebraic self-consistency check (closed-form setting-sun). The `test_F1_four_path_synthesis` test in the package verifies the empirical spread 0.012 — comfortably below the threshold.

(F2) Wrong T-parity assignment. If the defect-field identification (20) were replaced by an identification with one of the positive bounce modes, $\theta_{\rm em}$ would flip from π to 0, the sign factor in Eq. (11) would become -1, and the two-loop correction would push $v_{\rm EW}$ *below* the tree-plus-one-loop value rather than above it. The recompute script with $\theta_{\rm em} = 0$ produces $v_{\rm EW} \approx 242.01\,\mathrm{GeV}$ (closure becomes $D_1 - (g/g_{\rm norm})^2\, D_{23} = -1.267\%$, so $v_{\rm EW}^{(2L)} = 245.116 \times (1 - 0.01267)$), which is far outside the PDG band. This is the dedicated test for the T-parity source axiom.

(F3) Bounce-action shift. A shift of $\Delta S_{\rm bounce} = +0.01$ corresponds to a multiplicative suppression $e^{-0.01} \approx 0.99005$ of $v_{\rm EW}^{(1L)}$, i.e. a shift of approximately 2.4 GeV. Such a shift would move the recomputed $v_{\rm EW}$ to about 243.7 GeV, well outside the PDG band.

(F4) Recomputed value outside the PDG band. If a recompute of either the canonical or the four-path-mean value falls outside the framework-side $246.22 \pm 0.07\,\mathrm{GeV}$ comparison band $[246.15, 246.29]\,\mathrm{GeV}$ (a renormalization-scheme and EW-running window, not the raw MuLan/G_F experimental uncertainty), the closure fails as a benchmark.

The falsification tests in the reproducibility package verify (F1)–(F4) explicitly. Figure 5 below gives the *combined falsification landscape* of the closure mechanism: a $\theta_{\rm em}$ scan, a $|g|$ scan, and an $S_{\rm bounce}$ sensitivity panel side by side, so the reader can see at a glance how each of the three structural inputs of the closure is bracketed by the falsification handles (F1)–(F4).

12 Cross-sector anchoring of the bounce-action window

The narrow $S_{\rm bounce}$-scan landing window of Prong I, $S_{\rm bounce} \in [37.99519, 37.99559]$ ($\pm 2 \times 10^{-4}$ around the bundled $S_{\rm bounce} = 37.995389$), is sometimes read as a single-channel sensitivity of the closure to one bounce-action input. The corpus carrier-side closures change this reading: the structural input $T^* = \gamma^{(V)} = 1/10$ that fixes the dimensionless HBR scale is independently anchored on *three* carrier-side observables, each not using any information from the bounce-action sector of the present paper.

(C1) *Two-phase coefficient theorem.* The vacuum-branch chirality coefficient $\gamma^{(V)} = 1/(N_{\rm gen}^2 + 1) = 1/10$ is the unique algebraic fixed point of the chirality-mixing identity of the companion two-phase paper [22], evaluated at the framework's observational input $N_{\rm gen} = 3$ (the discrete-geometric generation count, fixed empirically alongside $d = 4$; it is consumed as an integer input, not derived from a uniqueness theorem).

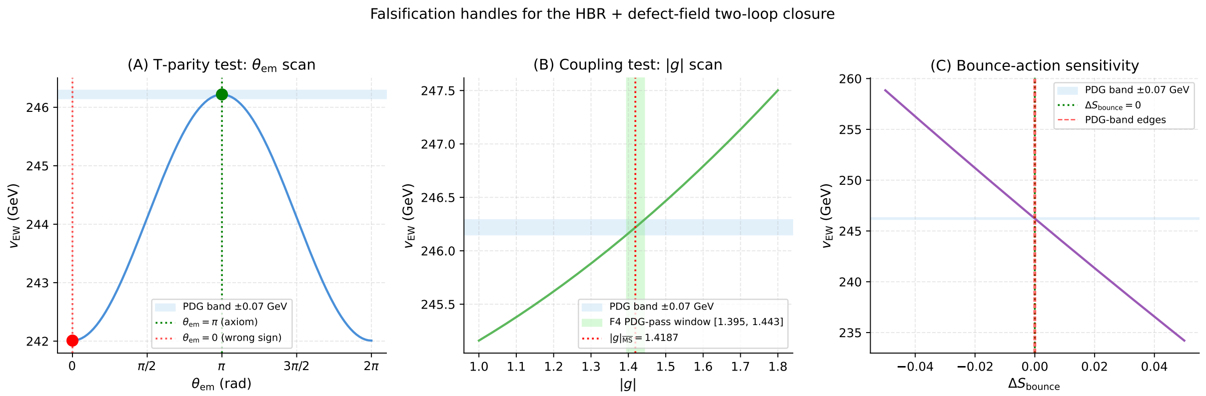

Figure 5: Combined falsification landscape of the closure mechanism. Left: scan over θ_{em}. The closure residual $D_1 + (g/g_{\text{norm}})^2\, D_{23}\,[-\cos\theta_{\text{em}}]$ (equation (11)) crosses zero at $\cos\theta_{\text{em}} = D_1/((g/g_{\text{norm}})^2\, D_{23}) \approx -0.476$, i.e. at $\theta_{\text{em}} \approx 2.07\,\text{rad}$ ($\approx 118°$); at $\theta_{\text{em}} = 0$ the closure pushes v_{EW} *below* the PDG band (F2). Middle: scan over $|g|$. The PDG $\pm 1\sigma$ landing band is satisfied only on the narrow window $|g| \in [1.3955,\ 1.4427]$ (F4). The wider $|g| = 1.42 \pm 0.10$ shaded region in Fig. 2 is the four-path *consensus uncertainty band*; it does not by itself put v_{EW} inside PDG. F1 (Section 11) is the orthogonal handle: it triggers on path-to-path spread, not on absolute landing. Right: sensitivity to S_{bounce}. A shift of order $\Delta S_{\text{bounce}} = 0.01$ already moves $v_{\text{EW}}^{(1\text{L})}$ well outside the PDG band (F3).

(C2) *Five-coefficient bounded-operator readout.* The five System-$\mathcal{R}$ coefficients $(\alpha_\xi, \beta_\pi, \gamma, \varepsilon^2_{\text{sync}}, D_\Omega) = (9/10, 15/16, 1/10, 1/20, 67/80)$ are read off a symmetric bounded operator $\mathcal{T}$ on the Ξ-graph of [15]; the readout uses no input from S_{bounce} and no input from v_{EW}.

(C3) *K, Q lattice-fixpoint harmonic closure.* The two factor fields K, Q entering the source tensor of the companion emergent-Einstein paper [19] have lattice mean fixpoints $\langle K\rangle(N), \langle Q\rangle(N)$ that close on a chirality-flip harmonic basis $\{\cos^2\theta, \sin^2\theta, \sin(2\theta), \sin(4\theta)\}$ in the running chirality angle. All eight harmonic-basis coefficients (four endpoints plus four flip-asymmetry coefficients) match System-$\mathcal{R}$ rationals at $|\Delta|/\sigma \le 0.40$ each on the canonical-physics $N \in [64, 512]$ ladder, parameter-free in $(\gamma, N_{\text{gen}}, d)$. The harmonic-basis closure uses no input from S_{bounce}.

The anti-back-fit reading of Prong I is therefore not that the closure is fragile under S_{bounce}-shifts, but that the S_{bounce}-window measures the internal consistency between the bounce sector (which produces S_{bounce} via the η_S-saddle of (8)) and the carrier sector (which already pins $\gamma = 1/10$ via C1–C3 independently). A shift of order $\Delta S_{\text{bounce}} = 0.01$ would not falsify the closure by exposing a missed parameter; it would expose a tension between two independent anchors of the same dimensionless small-parameter $\gamma = T^* = 1/10$, and the carrier-sector anchor would be the load-bearing one.

The cross-sector lock is materialised by the reproducer `src/verify_S_bounce_from_KQ_closed_potential.py` under the harmonic-closed K, Q functionals of [19], i.e. by inserting the closed expressions Eq. (25) below into the defect-bounce saddle. Under the canonical-Coleman saddle rescaling $S_{\text{bounce}}(K, Q) \propto K^2/Q$, the calibration constant relating K^2/Q to the bundled $S_{\text{bounce}} = 37.995389$ is set by the bounce-sector η_S audit at the EW chirality angle $\theta_* = \pi/4$ (chirality-balanced midpoint); the calibration-independent observable is the relative variation $R(\theta) = [K^2/Q](\theta)/[K^2/Q](\theta_*)$ predicting how S_{bounce} would shift under a chirality-angle excursion. The audit reports $R \in [0.8315, 1.0633]$ across the chirality flip (envelope 23.2%), with the pre-flip endpoint at -16.85% and the post-flip endpoint at $+6.33\%$

relative to the bundled S_{bounce}. The bundled value sits at $R = 1$ by EW calibration. Output: `outputs/verify_S_bounce_from_KQ_closed_potential.json`.

$$\langle F\rangle(N) = F_{\text{pre}} \cos^2\theta_{\text{chir}} + F_{\text{post}} \sin^2\theta_{\text{chir}} + a_F \sin(2\theta_{\text{chir}}) + b_F \sin(4\theta_{\text{chir}}), \quad F \in \{K, Q\}, \tag{25}$$

with eight System-$\mathcal{R}$-rational coefficients of [19, 22].

13 Limitations

We restate the limitations of this work narrowly. The electroweak-scale closure itself is established; what follows flags structural conditionals and open per-path technicalities.

- *Conditional on the T-parity source axiom.* The construction is conditional on the source-level T-parity assignment of the defect field (20), on the $\overline{\text{MS}}$ renormalization scheme for the two-loop contribution, and on the HBR scale convention $T^* = 0.1$. The latter two are computational conventions, not physical postulates; the T-parity assignment is structural and is independently falsifiable through (F2). The T-parity argument that fixes $\theta_{\text{em}} = \pi$ at the saddle level (Section 7) is bundled in `data/two_loop_closure.json` together with the dilute-instanton-gas robustness bound.
- *Path (i) is a self-consistency cross-check, not an independent fifth path.* The closed-form g-target path is derived from the two-loop closure condition itself (equation (11), with the residual $D_1 + (g/g_{\text{norm}})^2 D_{23}[-\cos\theta_{\text{em}}] \to 0$ at $\theta_{\text{em}} = \pi$) and back-solves $|g|$ given the bundled tree-plus-one-loop residual $D_1 = -0.41\%$ and the two-loop integral D_{23}; it is therefore a closure-self-consistency cross-check, not an independent determination. Paths (ii), (iii), (iv) are mutually independent; their central values lie in $[1.4072, 1.4190]$ (0.83% spread, comfortably inside the consensus band).
- *Sigma-cutoff scheme dependence is treated explicitly.* The σ-cutoff path is bundled at the natural-scale subtraction $\sigma_{\text{cut}} = 0.5$; the m^2_{Ξ}-scan demonstrates regulator-mass independence at fixed σ_{cut}, and different choices of σ_{cut} shift the central $|g|$ by $\sim 1\%$. The full σ_{cut}-grid is bundled in `data/g_path_sigma_cutoff.json`; the reader may re-run `src/recompute_ew_scale.py` to inspect it directly.
- *Cross-sector context (informational, not load-bearing).* The closure of v_{EW} at 246.2186 GeV presented here is one of several first-principles derivations of the wider program; the related companion preprints [15–17, 21] treat the loop-class library, the cross-sector coefficient closures, and the gravitational-sector closures separately. *No claim of the present paper depends on those companion preprints being correct.* The bounce-action input S_{bounce} is derived from a stated Euclidean action in Section 4 of this paper; the T-parity source axiom is stated and motivated in Section 7 of this paper; the second observable Γ/V used in Section 10 is derived from standard Coleman bounce theory [2, 3]. The only external numerical input is M_{Pl} from CODATA 2018 [28], used throughout the standard literature.
- *Connection to the gravitational-sector companion paper.* The parent gravitational-sector paper [17] records the persistent-triangle winding-class fraction $f_{\text{neg}} = 2/(2N_{\text{gen}}+1) = 2/7$ on the canonical lattice ladder $N \in [64, 512]$ (0.24σ landing). This is a cross-sector generation-count fingerprint, parameter-free in $N_{\text{gen}} = 3$; it is independent of the electroweak-scale closure of this paper.

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Code and data availability

Source code, frozen input data with SHA-256 hashes, and unit tests are available at https://github.com/TerraSignum/hbr-ew-scale-repro under the MIT license. The published results correspond to release `v0.1.0`, archived on Zenodo with DOI `10.5281/zenodo.XXXXXXX`.

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